

Particulate Matter

Week in Review

Week ending 3 October 2026 | 27 September 2026 to 3 October 2026 | 125 records pooled from 7 daily issues

125

RECORDS POOLED
(125 UNIQUE DOIs)

42%

MEASUREMENT SIDE
(52 RECORDS)

10

SUBTOPICS
ACTIVE

30

POOLED EFFECT
ESTIMATES (30 WITH A
CI)

28

ABSTRACT-FREE
(22% OF WEEK)

The week in one paragraph

Volume fell by nearly a quarter and the measurement side took it all back: 125 records, with *Exposure and Sensing* together at 42 % after a series-low 33 %. The abstract gap did not close — 28 of 125 (22 %) — and this rollup is being written six days late, because the pipeline stopped for four consecutive runs.

Scale. 125 records over seven issues on daily totals of 4, 15, 22, 25, 30, 15 and 14 — a $7.5\times$ range, by far the widest in the series, because Sunday 27 September returned **four**. Against recent rollups: 125 after 162, 148, 126, 150 and 109. All **ten** subtopics are populated for the second week running.

The measurement side recovered to 42 %. *Exposure assessment & modelling* 34 → 32 and *Sensing* 20 → 20 held in absolute terms while the total fell, so the share reads 54, 40, 44, 33, 42 % across five weeks. Last week's series low was a denominator effect, as this watch said at the time it might be; it was not a contraction of the instrumentation literature.

Occupational & indoor 22 → 14, and the indoor turn is still a model. The band that led the health side last week fell back to joint second with *Burden* (14 each). The week's indoor records are a school PM_{10} deposit with no abstract, waterpipe cafés at $285\ \mu\text{g m}^{-3}$, a stone-carving dust survey and a traffic-police ocular study — none of them an occupied-building intervention with a health endpoint. The AIRVAC protocol that entered the registry last week is still the only randomised test in view.

Abstract-free held at 22 %. 28 of 125, against 34 of 162 (21 %) — no improvement, and the retry pass named as the only fix in four consecutive rollups is **still not built**. 17 of the 28 sit on the measurement side, so the abstract gap continues to fall hardest on exactly the literature this watch exists to track. The Scholar Gateway connector, which is the designated recovery route for numbers abstracts omit, has been **down for the entire period** — first with an identity error, now ACCESS_DENIED.

Estimates: 30, every one with an interval. One estimate per 4.2 records (series 4.4, 2.7, 3.1, 5.2, 4.2), and 30 of 30 carry a CI — the second complete week in a row. Two issues (27 September, 1 October) contributed none.

Subtopic trend and inflection

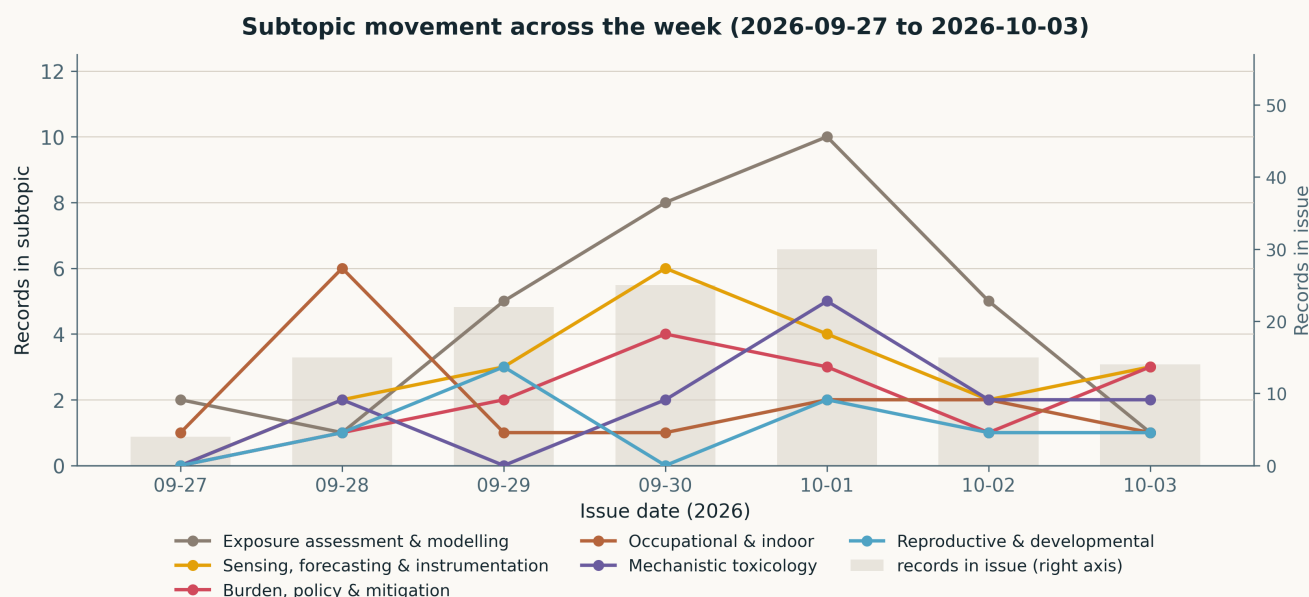


Fig. Per-issue subtopic counts across the week from `state/metrics.csv`, seven issues. Daily totals 4, 15, 22, 25, 30, 15 and 14 — a 7.5× range. The Sunday minimum (27 September, four records) is the smallest daily issue in the series; the 30-record Thursday (1 October) is a full weekday deposit. Five of the week's seven issues were built on a later run than their entry date, and the last of them — 3 October — was built **six days late**, on 9 October, after four consecutive runs halted during authoring. No gap was left.

Expanding. *Burden, policy & mitigation* 10 → 14 is the week's largest relative movement, carried by the coking-industry inventory, the Middle East conflict-emissions analysis, two GBD-type burden records and the Jilin CTM work. *Sensing* holds at 20 on a 25 % smaller week, which is a real gain in share.

Contracting. *Occupational & indoor* 22 → 14 (above) and *Mechanistic toxicology* 20 → 13 both give back most of last week's rise; *Cardiovascular & metabolic* 15 → 8 halves. *Respiratory* (10 → 5) is at a series low. *Reproductive & developmental* holds at 8 for the second week, and *Neuro* (8 → 9) and *Other clinical* (8 → 5) ease.

The inflection worth naming. Last week's black-carbon ordering was not tested, and the constituent question moved to nitrate instead. The test this watch set — whether a non-Chinese cohort reproduces BC leading the constituent weighting — went unanswered: no constituent-resolved cardiovascular cohort appeared this week in any country. What did appear, on the last day of the period, is the **China Kadoorie Biobank never-smoker lung-cancer analysis**: per IQR, **nitrate HR 1.35** (1.23–1.48) ahead of ammonium 1.22 and chloride 1.20, with the mixture at 1.13 and an additive nitrate × polygenic-risk interaction. Constituent-specific risk is now reported from three independent Chinese cohorts in three weeks, each naming a *different* leading constituent — BC for stroke and heart failure, nitrate for lung cancer. That is either real outcome specificity or collinearity in one shared reanalysis product, and the two are not yet distinguishable. **For a sensor watch the practical point is unchanged and sharpening:** nitrate is semi-volatile and reads low on heated and warm-cavity optical sensors, BC is measurable by low-cost absorption, and a mass-only network resolves neither.

Study architecture

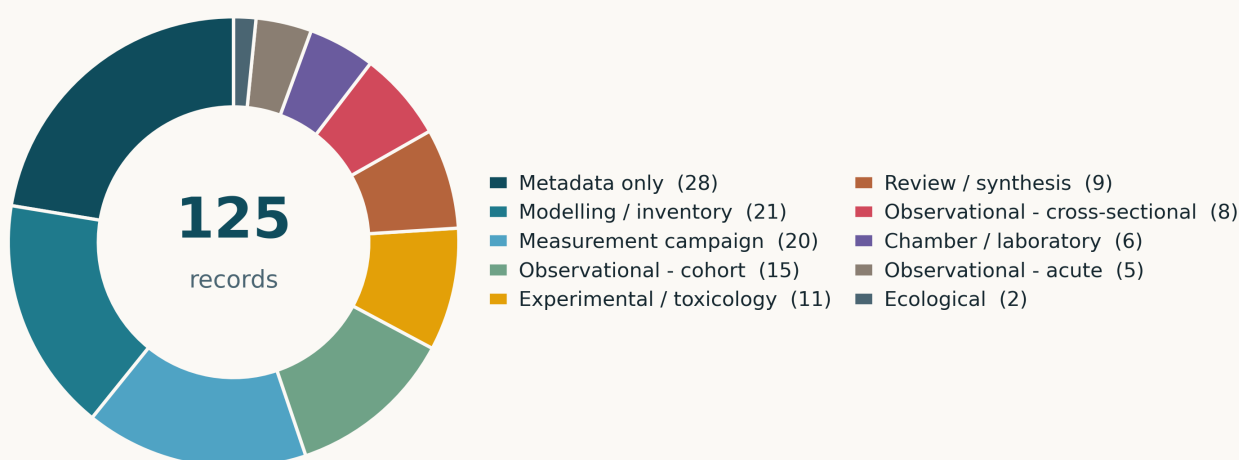


Fig. Study architecture over the pooled week. *Metadata only* leads again at 28, ahead of *Modelling / inventory* (21) and *Measurement campaign* (20) — the second consecutive week in which the missing-abstract bucket is the largest single group. Primary observational epidemiology (cohort 15, cross-sectional 8, acute 5, ecological 2) totals 30, down from 40. **No interventional record entered this week**, against four last week.

Pooled analytics

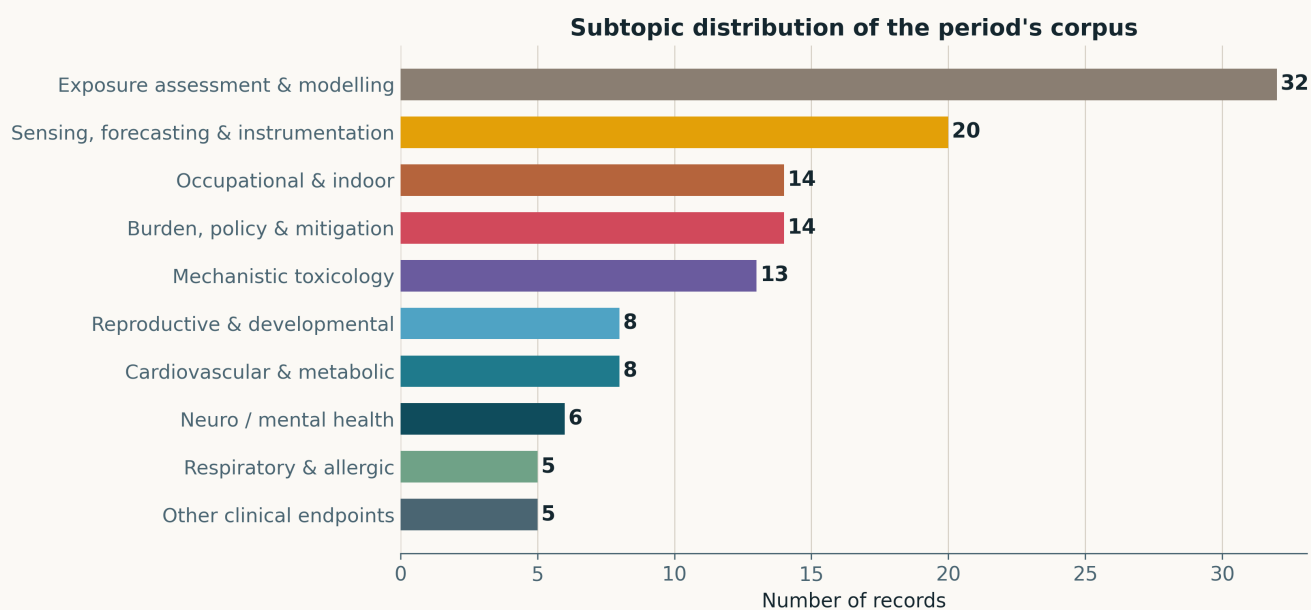


Fig. Pooled subtopic distribution, **all ten bands populated**. *Exposure* 32 and *Sensing* 20 put the measurement side at 52 of 125 (42 %). On the health side *Occupational & indoor* and *Burden* tie at 14, then *Mechanistic toxicology* 13; *Respiratory & allergic* at 5 is the weakest band of the week.

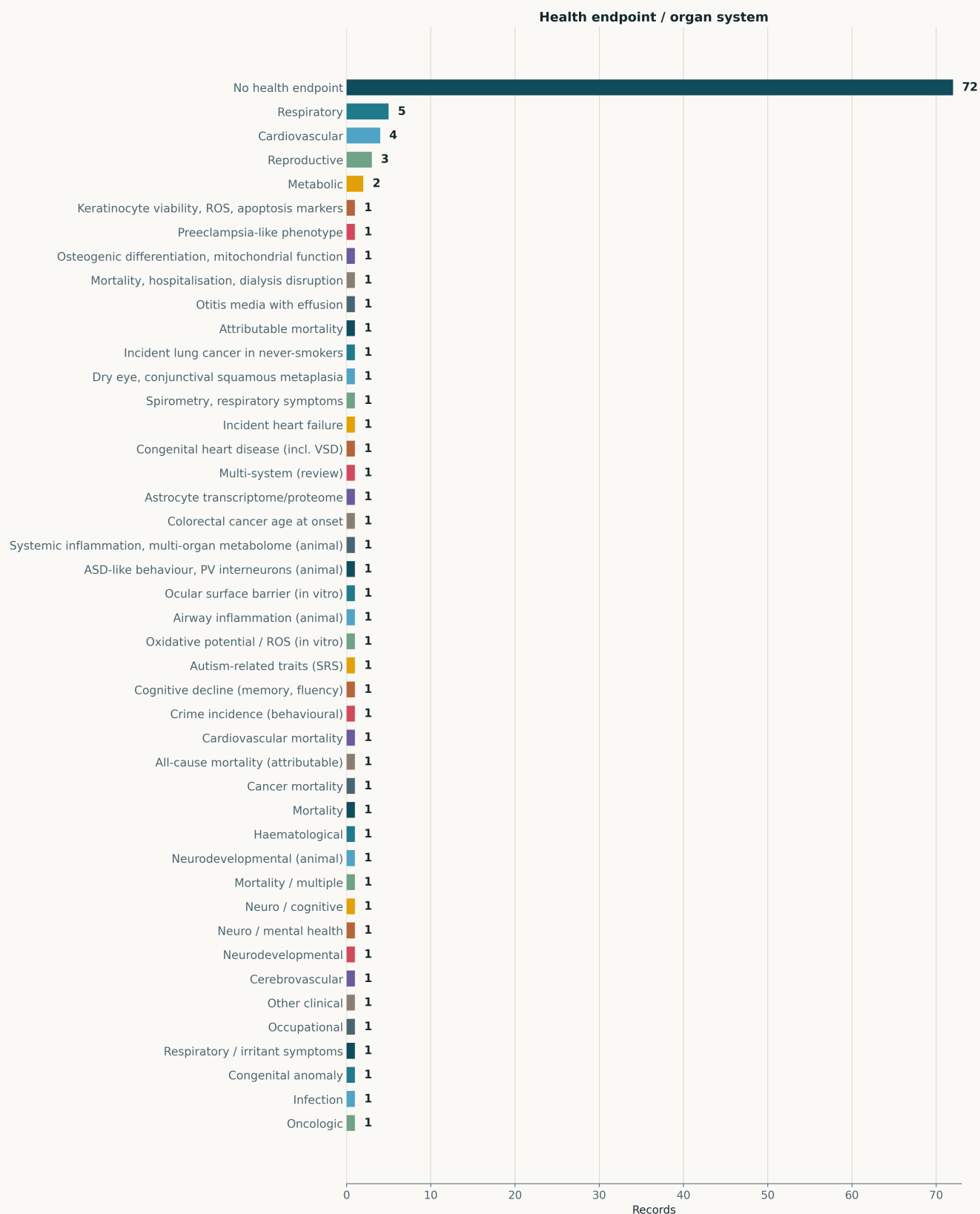


Fig. Health endpoint over the pooled week, canonicalised. “No health endpoint” pools the majority — 28 abstract-free records plus the exposure, instrument, emissions, optics and air-quality records. Among human outcomes, respiratory and cardiovascular lead at 4 each, then reproductive at 3 and metabolic at 2, with a long tail of single-record labels. A week in which no outcome label reaches five is a week with no clinical centre of gravity.

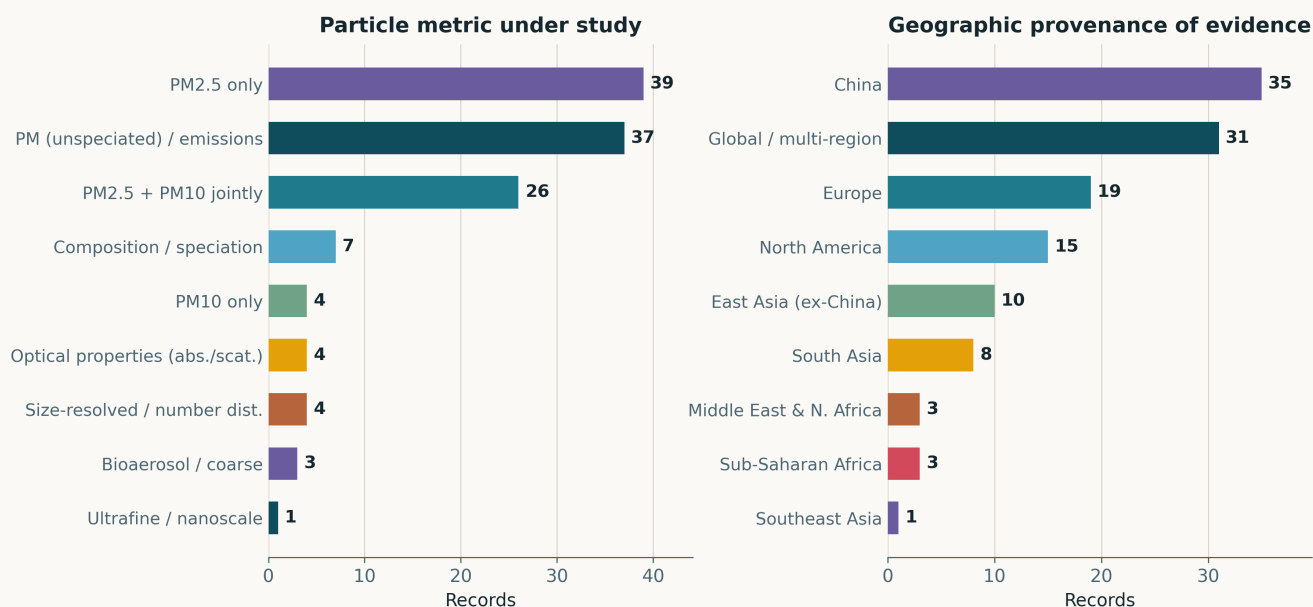


Fig. Left — particle metric. PM_{2.5} alone 39 of 125, unspeciated PM or emissions 37, joint PM_{2.5} + PM₁₀ 26 (a high share, driven by the indoor and occupational records that report both). The tail: **composition / speciation** 7, size-resolved 4, optical properties 4, PM₁₀ alone 4, bioaerosol 3 and **ultrafine** 1. Right — geography. *Global / multi-region* 31 again overstates reach, pooling reviews, laboratory work and the 18 metadata-only records whose country is not recoverable. **China** 35 is the largest single-country bar — 28 % of the week — then Europe 19, North America 15, East Asia (ex-China) 10 and South Asia 8; Sub-Saharan Africa fell to 3.

Life-course windows addressed this period (records may span several stages)

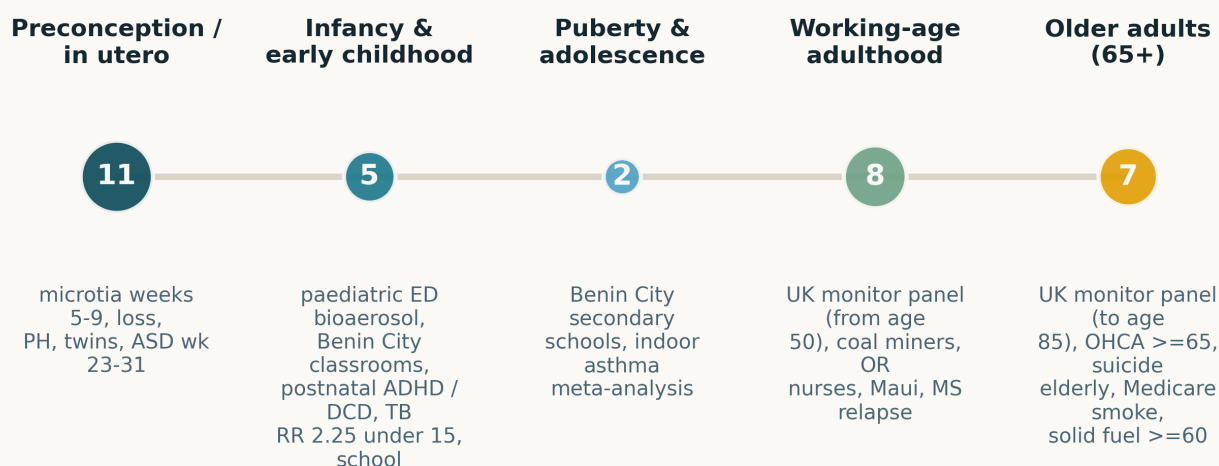
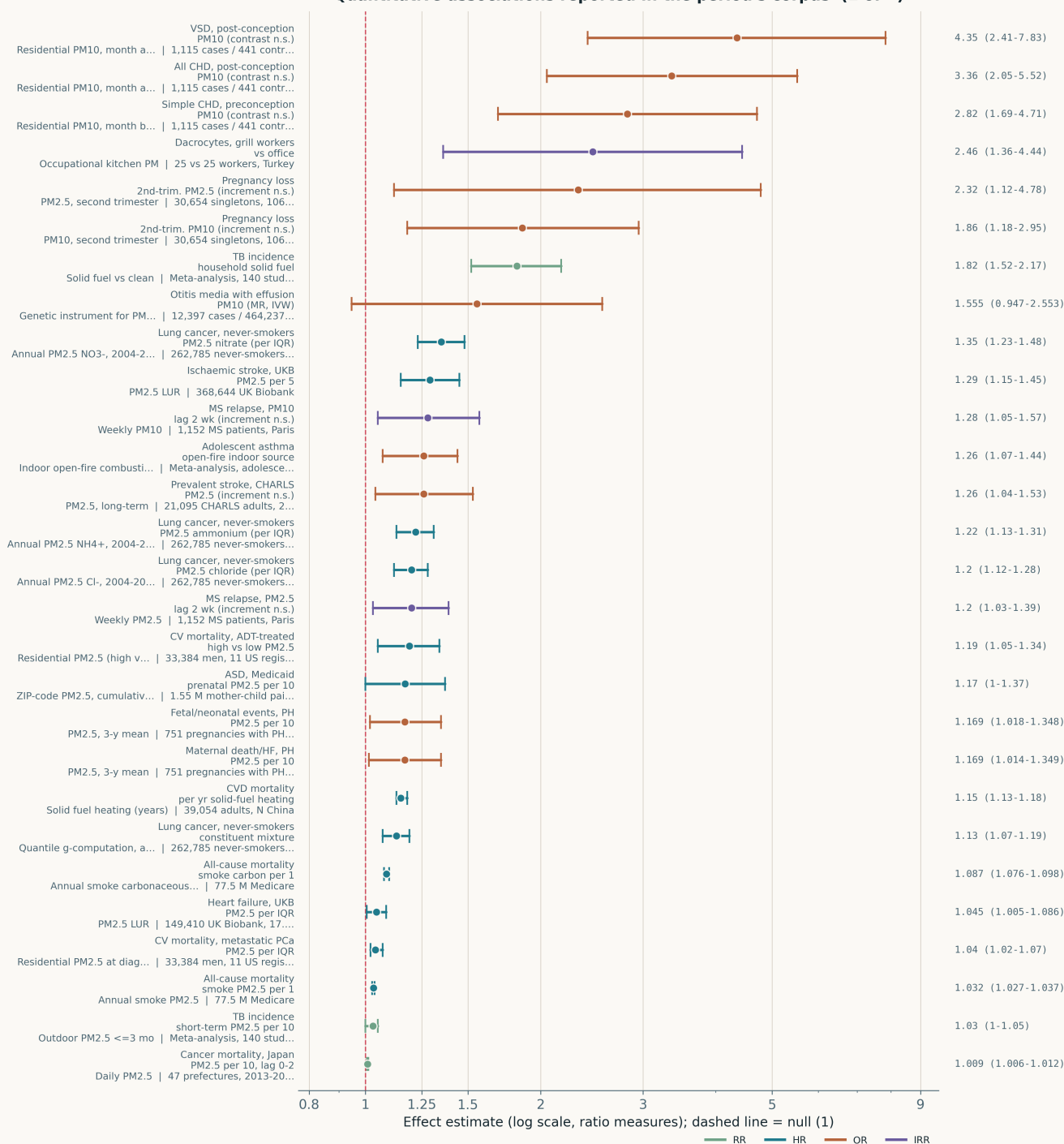


Fig. Life-course windows across the pooled week; counts non-exclusive. **Preconception / in utero** 11 leads for the first time in the series, carried by the Delhi congenital-heart case-control, the pregnancy-loss and pregnancy-hypertension cohorts and the Medicaid prenatal autism analysis. Working age 8 and older adults 7 follow; infancy 5 and **adolescence** 2 — the weakest window for the sixth consecutive week.

Quantitative associations reported in the period's corpus (1 of 2)



Quantitative associations reported in the period's corpus (2 of 2)

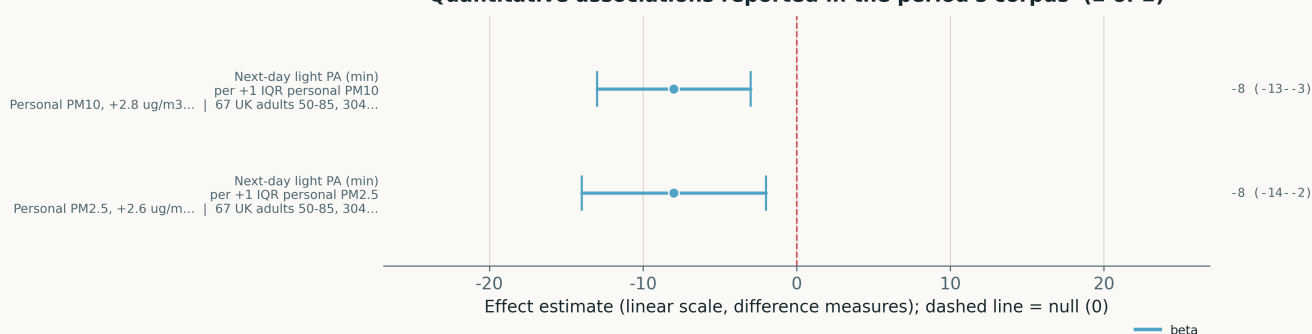


Fig. All 30 pooled estimates, ratio and difference scales in separate panels. **Not mutually comparable and not meta-analysed:** contrasts include per-IQR, per-SD, per-10 and per-1 $\mu\text{g m}^{-3}$, per-year, dichotomised and — in four cases — unstated increments, on HR, OR, RR, IRR and β scales; heterogeneity is structural, so **no I^2 is reported** and none would be meaningful. Readable points: **the three largest ratios are all from one case-control** — the Delhi congenital-heart ORs of 4.35 (2.41–7.83), 3.36 and 2.82, on 1,115 cases against 441 controls with the PM₁₀ increment unstated, which is why they sit so far right of everything else; **the tightest intervals are the Medicare smoke-PM mortality pair** (1.032, 1.027–1.037 and 1.087, 1.076–1.098) on 77.5 million beneficiaries; the four China Kadoorie

constituent estimates share cases and are nested, not independent; and the two-sample Mendelian randomisation OR for otitis media (1.555, 0.947–2.553) is the only estimate on the panel whose interval crosses unity by design — the authors report it as a bounded null. The difference panel carries the two next-day physical-activity β coefficients from the 67-adult UK personal-monitoring panel (–8 minutes per IQR, both PM_{2.5} and PM₁₀).

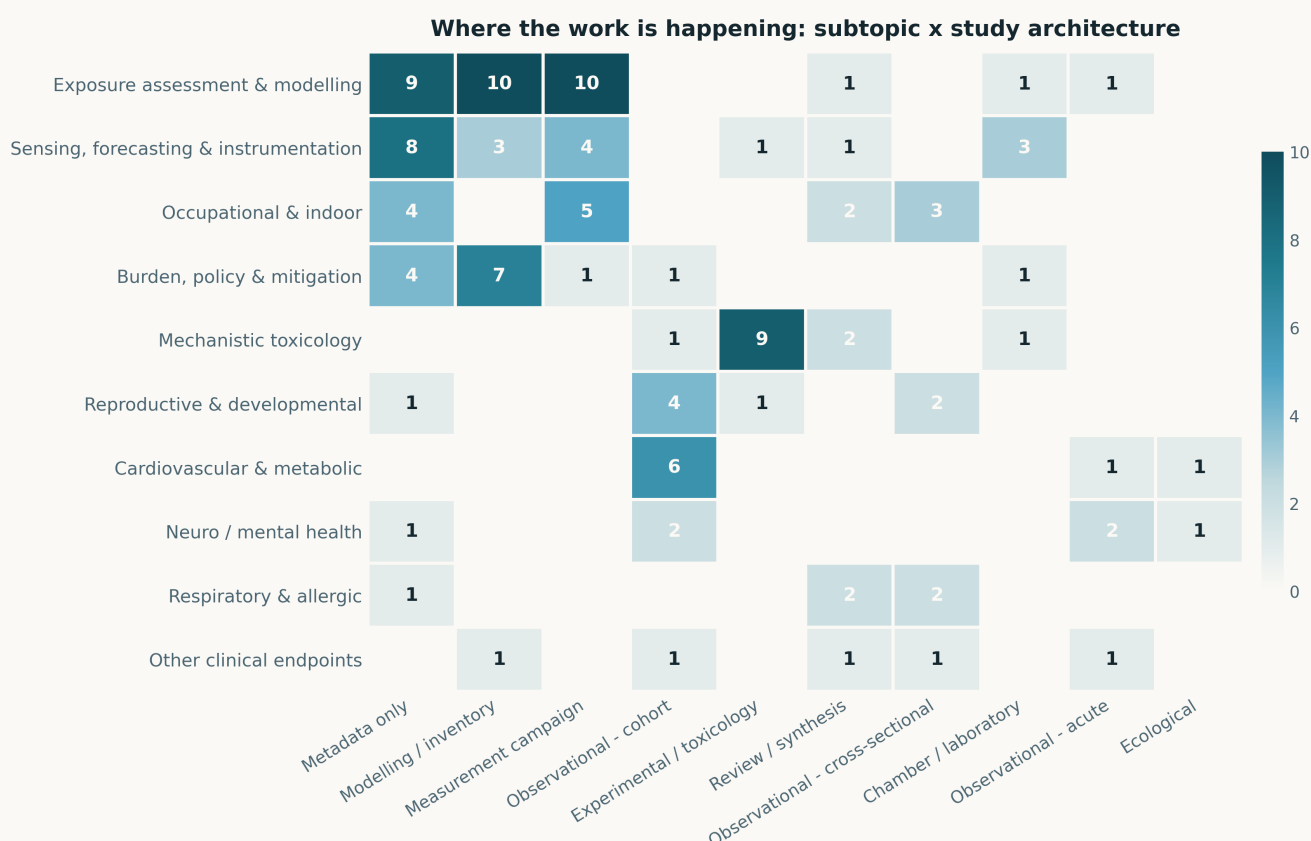


Fig. Subtopic $\times$ architecture, pooled. The densest cells are *Exposure $\times$ Metadata only* and *Exposure $\times$ Modelling*, then *Sensing $\times$ Metadata only* — the abstract gap is again concentrated on the measurement side, where 17 of 52 records (33 %) carry no abstract. The standing finding survives a tenth week: of the 52 measurement-side records, 2 carry any outcome label and none observes a human outcome — one is a modelled carcinogenic risk index, the other an emissions-policy label.

Venues, trial watch, and what the pipeline did

Venues. A two-way tie at 9 — *Environ Sci Technol* and *Atmos Environ* — then *Atmosphere* and *Indoor Air* at 7, *Front Public Health* and *J Environ Sci (China)* at 6, *Atmos Pollut Res* at 5. The leading venue holds 7 %. *Atmos Environ* and *Atmos Pollut Res* remain the main sources of metadata-only records (Elsevier same-day deposits), exactly as in the last three rollups. **Preprints supplied 11**, up from 6 and the highest of the series — seven of them medRxiv.

Recurring authors. Unchanged: surname frequency is a naming artefact, and **no repeat-group cluster is claimed**. No dataset repeated three times this week as CHARLS did last week; the China Kadoorie Biobank and UK Biobank each supplied one record.

Trial watch — nothing new in the period. The ClinicalTrials.gov leg was queried on each run covering the week (LastUpdatePostDate windows 26–29 September, 30 September, 1 October and 2–3 October) and returned **one hit with nothing kept**: NCT07850648 (CAN-EXACT, McGill, observational COPD-exacerbation cohort, $n=125$, not yet recruiting), whose `analyze_endpoints` shows primaries of exacerbation rate and diagnostic markers with environmental factors only inside a generic secondary risk-factor measure — **no PM or air-quality measurement**, so it was screened and not added. Carried from the previous period: NCT06541691 (COATED-AIR) still reads *recruiting* past its listed primary completion and still registers **no exposure or adherence endpoint**; NCT02944682 (HAPIN) posted a registry update. `trials.json` holds **16 trials** over **27 logged windows**; the 16-vs-24 count discrepancy flagged in the August monthly was **audited and closed** on the 2 October run — it was a drifted tally in `counter_note`, and all counters were re-derived.

What the harvesting legs did. Crossref-by-ISSN was the highest-yield leg every weekday (40–78 records) and is also the source of most of the week's missing abstracts. Europe PMC ran clean all week after the 25 September outage (17–75 records/day). The harvester's PubMed sensing query returned 0–3 daily, and the PubMed connector remained load-bearing, appending 2–7 records a day that the two harvester queries missed. OpenAlex returned HTTP 429 on every single run of the period — it has now contributed nothing for three weeks and should be treated as dead rather than degraded. arXiv answered (148 kB) but had no entry dated inside the window on any day. Consensus returned 10 calibration hits and 0 new records every day, for the sixth consecutive week. Scholar Gateway failed on every attempt, first INVALID_QUERY: could not resolve user identity, later ACCESS_DENIED; it needs the user to reconnect it. Two real defects were found and fixed inside the period: the month-boundary gate in `_screen_0929.py` that was discarding end-of-month preprints on a 1st-of-month issue (5 records recovered), and the batch-free-month regime in `plots_weekly.py` that had shipped a stale legend in the August monthly.

Open gaps

- ▶ **The pipeline stopped for four consecutive runs, and this rollup is six days late.** The 6, 7, 8 and 9 October runs harvested and screened but did not author; three of them halted part-way through writing a single large issue file. The 3 October daily and this weekly were both recovered on the 9 October run, and authoring is now **chunked** (`authoring/<date>/p*.py` plus `_mk_chunked.py`) so that a partial run leaves usable work behind. This is the largest availability failure in the series and the defect most worth watching next week.
- ▶ **The abstract-retry pass, fourth week running.** 22 % of the week shipped without an abstract and **17 of those 28 are measurement-side records**. The highest-relevance titles are again among them — the Central Himalaya CUPi sensor network, a composition-to-volatility parameterisation, Kuwait PM episode statistics, a size-resolved dry-deposition chamber study. With **Scholar Gateway down for the whole period**, the pipeline currently has no working recovery route for numbers an abstract omits.
- ▶ **Constituent-specific risk is now reported from three Chinese cohorts naming three different leading constituents.** Outcome specificity and reanalysis collinearity are not yet distinguishable, and no non-Chinese constituent cohort has appeared. The test stands.
- ▶ **None of 52 measurement-side records observes a human outcome**, tenth consecutive week — and this week not even one, against one last week.
- ▶ **Ultrafine and size-resolved number: 5 of 125.** The week's UFP-relevant work is entirely atmospheric; no health record used a number concentration.
- ▶ **Pipeline defects carried forward:** no first-author affiliation field; the Cancer/Oncologic split in `ENDPOINT_CANON`; no `\DIGESTDATE-vs--date` build guard; the 2026-07-27 corpus missing `LIFECOURSE`; `.git` lock debris accumulating on the delete-restricted mount. `metrics.csv` continues clean (`csv.writer`, 541 rows at period close, 0 malformed).

Paper-level digest — the week by subtopic

Every record that appeared in a daily issue keeps that issue's critical summary verbatim; nothing here is re-written or re-assessed. Records surfaced without an abstract are shown as such. Ordered by subtopic, then by the date of the issue that first carried the record.

Sensing, forecasting & instrumentation

20 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (20 OF 20 RECORDS IN THIS SUBTOPIC)

Aakko-Saksa et al. 2026 Environ Sci: Atmos

Aggregated non-methane organic and inorganic gases as indicators of aged particulate matter induced by traffic

METADATA ONLY. Metadata-only record: RSC deposited title, authors and DOI on 1 Oct with no abstract; PubMed efetch, Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The deposit carries a one-line teaser: aggregate NMVOC emissions were found to indicate the organic fraction of the subsequently aged PM. If it holds, a gas-phase total (cheap to sense) could flag secondary organic PM that optical counters see only after ageing; the data and the lag structure are unknown. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1039/d6ea00024j

Chemyakina et al. 2026 Opt Lett

Single-scattering look-up table for remote sensing and in situ black carbon studies

Tabulates single-scattering properties of fractal-like soot aggregates (refractive index 1.95-0.96i; fractal dimension 1.4-2.8; 1-4,096 monomers of 15-25 nm radius) across 0.35-2.3 μm for PACE retrievals, giving a speed-up of at least 10^6 over direct computation. As a test it reproduces the low single-scattering albedo of in situ ACTIVATE aircraft aerosol. Limitation: bare, identical, touching monomers only - no coating, no polydispersity, no necking - so aged, internally mixed BC (the ambient case) and its absorption enhancement fall outside the table; agreement with one campaign is a plausibility check, not validation. Relevance: moderate - fast BC optics is what multi-wavelength absorption channels on low-cost nodes need to convert attenuation to mass with a defensible morphology assumption.

doi: 10.1364/ol.609835

Chen et al. 2026b J Aerosol Sci

Linking Aerosol Chemical Composition to Volatility: A Predictive Parameterization and Implications for Future Clean Air

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 3 Oct without an abstract. By title, a parameterisation predicting aerosol volatility from chemical composition. Volatility is the quantity that decides how much of an ambient aerosol survives the heated inlet of a reference monitor or the warm optical cavity of a low-cost sensor, so a composition-based predictor is directly relevant to the semi-volatile bias that separates sensor and FEM mass. Relevance: potentially high; on the retry list.

doi: 10.1016/j.jaerosci.2026.106902

Hall et al. 2026 Indoor Air

Indoor Air Quality Forecasting With Machine Learning: A Systematic Review

Systematic review of 99 studies (2010-2026) on machine-learning forecasting of indoor air quality. Output grew sharply after 2020; LSTM-type time-series models dominate and transformers are emerging for longer horizons. Only 13% of studies use occupancy or behavioural inputs, although occupant activity drives indoor peaks. Evaluation is almost entirely RMSE and R2, with little assessment against health-relevant thresholds or by forecast horizon, and Africa and South America are barely represented. No pooled skill metric is given, so the review maps practice rather than grading it. Relevance: high for indoor sensor networks - it is the same validation gap Sahin et al. show today: headline R2 without horizon-specific or peak-event skill says little about operational usefulness.

doi: 10.1155/ina/8524332

Herman et al. 2026 Atmos Environ

An Advanced, Multifunctional, and Extreme-Condition Test Chamber for Performance Evaluation of Air Quality Sensors

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes a chamber for evaluating air-quality sensors under extreme conditions. Directly relevant to LCS characterisation: humidity, temperature and high-load excursions are where field calibrations fail. First on the retry list. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122401

Hu et al. 2026 J Environ Sci (China)

Columnar optical properties of locally emitted light-absorbing carbonaceous aerosols

Paired surface and mountain-station aethalometer (AE33) measurements in Beijing under pollution controls; an ensemble HYSPLIT + concentration-weighted-trajectory screen isolates locally emitted events, and a wavelength-dependent AAE method derives BrC absorption at 370 and 470 nm. BrC falls off more steeply with height than BC, so the BC share of absorption rises aloft. During local events, noon AOD and AAOD rise 1.91- and 1.93-fold and SSA shows a noon minimum. Limitation: columnar optics are reconstructed from two point sites plus Mie theory for scattering, not measured by a sun photometer or lidar, and trajectory screening of 'local' is model-dependent. Relevance: moderate - surface BrC/BC ratios from ground nodes do not transfer to the column that satellite fusion uses.

doi: 10.1016/j.jes.2026.02.005

Jonsson et al. 2026 Atmos Chem Phys

Tracing biological, anthropogenic, and inorganic sources of coarse aerosols via single-particle fluorescence and optical morphology

Reference library of fluorescence spectra and morphology for coarse-mode particles (pollen, dust, bacteria, microplastics) generated in controlled experiments on a Multiparameter Bioaerosol Spectrometer, then used to train a supervised classifier tested against chemical tracers at Zeppelin Observatory, Svalbard. Biological and non-biological particles overlapped enough to bias fluorescence-only classes; the classifier confused biomass burning with biological particles and dust with sea spray. Domain adaptation with co-located observations reproduced the published annual bioaerosol cycle and raised summer concentrations toward offline values. No per-class accuracy is given in the abstract, and lab aerosols may not represent ambient ones. Relevance: moderate - an open classifier for real-time coarse-particle typing, the fraction optical LCS mass-convert worst.

doi: 10.5194/acp-26-13617-2026

Kang et al. 2026 Build Environ*Moisture-coupled aerosol filtration in moss-based porous media: enhanced particle capture and aerodynamic resistance trade-off*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 3 Oct with no abstract. By title, a filtration study of moss-based porous media in which moisture improves particle capture at a cost in aerodynamic resistance. The efficiency-vs-pressure-drop trade-off is the same constraint that governs inlet and filter design in gravimetric samplers and in filtered-air intervention trials. Relevance: low-moderate; on the retry list.

doi: 10.1016/j.buildenv.2026.115363

Mathissen et al. 2026 Atmosphere*Development of the Worldwide Harmonized Heavy Duty Brake Cycle (WHBC) for Brake Wear Particle Measurement*

Builds a regulatory brake-wear test cycle for heavy-duty vehicles (Euro 7 covers HDV non-exhaust PM but GTR 24 has only a light-duty cycle). Brake events were extracted from a global activity dataset and matched on empirical distributions of key brake metrics: 246 events over 271 km and 5 h 39 min. CO₂-monitoring data imply ~73% of kinetic energy is dissipated by friction brakes rather than retarders and drag. Dynamometer runs show the cycle is feasible and repeatable; particle results are described only as first insights, so no emission factor is reported. Limitation: retarder and regenerative braking share will shift with fleet electrification. Relevance: moderate - harmonised non-exhaust emission factors are what source-apportionment kerbside sensor signals.

doi: 10.3390/atmos17100964

Morel et al. 2026 Indoor Air*Evaluation of Two Cyclonic Bioaerosol Samplers for Hospital Air Monitoring in a Calibrated Test Chamber Integrating Aerodynamics and Biological Endpoints*

Coriolis+ and Coriolis Compact cyclones were challenged in a calibrated 20 m³ chamber with infectious SARS-CoV-2, *S. aureus* and *A. fumigatus* at several concentrations, scored on genomic and viability endpoints, and paired with CFD and airflow measurements. Neither detected infectious virus at the lowest level; Coriolis+ recovered nearly all viral genomes and more viable bacteria and fungi, while the Compact recovered more genomes of the two non-viral organisms at moderate-high loads. Ranking flips with endpoint, so 'collection efficiency' is not one number. No quantitative recoveries in the abstract. Relevance: low-moderate - bioaerosol, not PM mass, but a reference protocol for any sampler co-location.

doi: 10.1155/ina/2773156

Rawat et al. 2026 Atmos Environ*Integrated Assessment of PM_{2.5} in the Central Himalaya using Satellite, Reanalysis, and a new In-situ CUPI Sensor network*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 3 Oct with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still refusing the connection. By title this is exactly the fusion problem this watch tracks - a new in-situ sensor network (CUPI) evaluated against satellite and reanalysis PM_{2.5} in complex mountain terrain, where both gridded products are weakest. Relevance: potentially high; on the retry list, and worth a full-text read when the abstract appears.

doi: 10.1016/j.atmosenv.2026.122395

Rosca & Stancu 2026 Indoor Air*Indirect Machine Learning Prediction of a Composite Indoor Air Quality Index in an Airport Passenger Transit Area*

A composite index (PIAQI) built from indoor CO₂, PM_{2.5} and PM₁₀ at Brindisi Airport is predicted without those three inputs, from other indoor aerosol channels, weather and public outdoor pollution data. Seven regressors were trained on 1 million high-frequency observations; XGBoost was best. Validation is a random 70/30 split of an autocorrelated series - the design Sahin et al. showed in the 29 Sep issue inflates R² - and the 'alternative aerosol measurements' retained are themselves close proxies for the excluded PM channels, so the task is closer to interpolation than to indirect sensing. Relevance: low - an example of the leakage-prone validation practice the field needs to retire.

doi: 10.1155/ina/3119205

Sahin et al. 2026 Atmosphere*Validation Design Governs the Apparent Performance of Machine Learning Calibration for Low-Cost Optical Particle Counters: A PM₁ Study*

Three low-cost OPCs were co-located with reference nephelometers over seven indoor runs of 20-33 h (Jun-Sep 2022; reference PM₁ 8.4-9.2 μg m⁻³). Factory output under-read by 5.3-6.1 μg m⁻³ (RMSE 6.5-7.6, 74-82% of the mean). Six calibrators (MLR, SVR, RF, GB, XGBoost, LightGBM) reached R² 0.86-0.89 under a random 80/20 split, but R² collapsed under day-blocked and run-blocked splits and sat near or below zero under forward chaining and chronological hold-out. Calibration still cut prospective RMSE 22-48%, yet the calibrated slope was 0.18-0.44, compressing real variation. Validation design explained 63-91% of the R² spread, algorithm choice 3.6-10.6%. Limits: three sensors, ~200 h, narrow indoor range. Relevance: direct - random-split R² in LCS calibration papers is leakage from autocorrelation; require temporally blocked validation.

doi: 10.3390/atmos17100947

Tu et al. 2026 Indoor Environ*Assessing the Need for Site-Specific Calibration of ATP Bioluminescence Sensors for Airborne Bacteria Monitoring in Indoor Environments*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title asks whether ATP-based airborne-bacteria sensors need site-specific calibration. The same transferability question PurpleAir regional-calibration work asks for optical PM sensors. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.indenv.2026.100197

Wang et al. 2026 npj Clim Atmos Sci*Artificial intelligence for atmospheric chemistry: from observation to process understanding*

METADATA ONLY. Metadata-only record: Springer Nature deposited title, authors and DOI on 28 Sep with no abstract; PubMed efetch, Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates a review or perspective on AI from atmospheric observation to process understanding. Of interest only if it covers sensor-network calibration or PM retrieval. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1038/s41612-026-01559-6

Xu et al. 2026 Environ Sci Technol*Integrating Fragmented Mobile Monitoring Data into a High-Resolution Spatiotemporal Exposure Model for Black Carbon*

Vehicle-borne eBC measured in three seasonal campaigns (2023-2024) across residential Beijing, aggregated hourly, then gap-filled in time with XGBoost on external temporal predictors (regulatory pollutants, meteorology) and interpreted with SHAP; site-day estimates feed a hierarchical geostatistical model applied to pregnancy-cohort addresses. XGBoost reached 10-fold CV R² 0.69 against site-hour eBC; the spatial stage is reported only as internal interpolation at monitored points, so there is no external validation at unmonitored cohort homes - the step the cohort actually uses. Two stacked models also propagate error the paper does not quantify. Relevance: high - the same temporal-gap problem affects any campaign- or sparse-node design, and the two-stage template transfers directly to low-cost BC/PM networks.

doi: 10.1021/acs.est.6c08120

Yan et al. 2026 Atmosphere*Baseline Determination for Threshold-Based Interpretation and Groups of Single-Particle Aerosol Fluorescence Measurements*

Tests deionised water, NaCl and SiO₂ as low-fluorescence baselines for the SwisensPoleno Jupiter and compares $\mu+3\sigma$, P99 and MAD thresholds. NaCl gives the broadest baseline ($\mu+3\sigma$ upper thresholds $\sim 52.0, 39.3, 38.2 \times 10^{-3}$ a.u. across excitation sources, vs $6.6/10.1/3.1$ for H₂O and $12.1/9.1/2.6$ for SiO₂), and the effect persists within a shared 10-30 μm size range; baseline choice changes the fraction of fluorescence-positive particles in both test materials and summer/winter ambient data. P99 tracks $\mu+3\sigma$; MAD diverges. Limitation: one instrument model, and no ground truth for which threshold best recovers biological particles. Relevance: moderate - fluorescence-positive counts from different networks are not comparable unless baseline material and threshold rule are reported, which is a metadata standard LCS bioaerosol channels lack.

doi: 10.3390/atmos17100971

Yang et al. 2026 Sensors (Basel)*An NB-IoT-Based Architecture with Spatial-Statistical Analytics for Cross-Domain Air and Water Quality Monitoring in Aquaculture*

Edge nodes with low-cost air (PM₁/2.5/10, O₃, TVOC, eCO₂, T/RH, UV) and water sensors stream over NB-IoT and MQTT to a cloud database with IDW mapping and alerting. A deployment at one aquaculture facility logged 14,400 one-minute records with 99.24% raw packet delivery; analyses correct for autocorrelation ($N_{\text{eff}} \sim 892$) and FDR, and the headline findings are air-temperature/dissolved-oxygen and humidity/turbidity correlations. Limitation for this watch: no PM sensor is co-located with a reference, calibrated or humidity-corrected, so the PM channel is uncharacterised; the PM data enter none of the reported results. Relevance: low - the telemetry (gateway-free NB-IoT, packet delivery) is the transferable part; it says nothing about PM data quality.

doi: 10.3390/s26185964

Yang et al. 2026 Environ Sci Technol Lett*Development and Diagnosis of a Temporal-Context U-Net Deep Learning Surrogate Model for CMAQ Particulate Nitrate over China*

An attention U-Net takes the target hour, the previous 2 h and a 96 h summary to emulate CMAQ surface nitrate over China; it beats baseline surrogates, especially at high concentrations. Emission-reduction tests reproduce the direction of July national-mean nitrate decreases but underestimate their magnitude, and agreement is poor for January NO_x responses outside training months. Residuals correlate with CMAQ NH₃ and particle H⁺, and adding them helps. Limitation: it emulates a model, not the atmosphere - CMAQ's own nitrate biases are inherited - and the weak emission-response skill is exactly what a policy surrogate needs. Relevance: moderate - nitrate is the PM component optical sensors misread most under humidity, and fast surrogates are a route to composition-aware calibration.

doi: 10.1021/acs.estlett.6c00755

Zhou et al. 2026 Aerosol Sci Technol*Effects of geometrical parameters on eccentric cylindrical differential mobility analyzer*

METADATA ONLY. Metadata-only record: Taylor & Francis deposited title, authors and DOI on 28 Sep with no abstract; PubMed efetch, Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates a study of how geometry affects the performance of an eccentric cylindrical differential mobility analyser. Relevant to compact mobility sizers - the reference against which low-cost ultrafine and size-resolving sensors are characterised. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1080/02786826.2026.2734523

Exposure assessment & modelling**32 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (32 OF 32 RECORDS IN THIS SUBTOPIC)

Akhtar et al. 2026 Environ Monit Assess*Spatiotemporal characteristics of ozone and PM_{2.5} pollution in Jilin City: insights from observations in 2020 and 2023*

WRF-CMAQ at 3 km plus surface observations compare May-July 2020 (COVID low-emission baseline) with 2023 in a petrochemical city in Northeast China. Meteorology explains $\sim 35\%$ of O₃ variability (temperature-dominated) but $<1\%$ of PM_{2.5}; PM_{2.5} rose in May and fell sharply in July, staying high in Huadian, Chuanying and Jiaohe. Limitation, stated by the authors: emissions and meteorology both changed between the two years, so attribution is associational, and O₃ is systematically underestimated; $<1\%$ explained PM_{2.5} variance means the chosen predictors simply do not describe PM_{2.5} here. Relevance: low - mid-size industrial cities are under-monitored, but this adds two seasons, not a method.

doi: 10.1007/s10661-026-15929-3

Barthaux & Zuend 2026 Environ Sci: Atmos*Estimating Speciated Input Concentrations for Biomass Burning Organic Aerosol Modeling*

METADATA ONLY. Metadata-only record: RSC deposited title, authors and DOI on 1 Oct with no abstract; PubMed efetch, Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The deposit carries only an opening sentence on biomass burning VOC and primary organic aerosol inputs for increasingly complex chemical models. Relevant to smoke-PM modelling, which the 30 Sep issue showed carries the mortality excess. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1039/d6ea00135a

Bi et al. 2026 J Environ Manage*Associations of street morphology and local wind environment with annual cumulative ambient PM_{2.5} exceedance: Evidence from Wuhan*

6,785 Wuhan streets in five canyon typologies; the outcome is annual cumulative PM_{2.5} exceedance. CatBoost-SHAP and PLS path models explained $>70\%$ of variance on average; geometry plus wind contributed $>50\%$. Enclosed canyons were governed by Building View Index and Sky View Factor; in open canyons street-wind angle and speed contributed 25% on average. Wind-speed breakpoints: 0.33 m/s (95% CI 0.31-0.44) enclosed, 1.19 m/s (0.78-1.31) open. Visible greenery was positively associated with exceedance in enclosed canyons, canopy ratio negatively. The street-level PM_{2.5} field is modelled, not measured, so the typology effects inherit the exposure model's structure. Relevance: moderate - canyon type and wind thresholds are a siting stratification for street sensors.

doi: 10.1016/j.jenvman.2026.130987

Blond et al. 2026 Preprints.org (preprint)

Boundary-Layer Height Outweighs Wind Speed in the Meteorological Control of Urban Air Pollution: Unique-Variance Attribution and a City Typology for Ten French Cities

Semi-partial R² separates the unique variance of boundary-layer height and wind speed for PM_{2.5}, PM₁₀, NO₂ and O₃ in France's ten largest cities (2018-2025), across pooled, deseasonalised and detrended frames, checked with GAMs. BLH and wind correlate at $r \sim 0.70$ (0.81 on anomalies); on pooled levels BLH carries four (NO₂) to nine (PM) times the unique variance of wind, a lead robust for PM. A three-type city typology (ventilation-, transitional, BLH-controlled) predicts which covariate a city needs. BLH is reanalysis-derived, not measured, and the preprint is not peer reviewed. Relevance: high for sensor-network trend normalisation and short-term epidemiology that adjust for wind alone - BLH should be a first-tier covariate for PM.

doi: 10.20944/preprints202609.2417.v1

Bloomberg et al. 2026 medRxiv (preprint)

Associations of individual exposure to particulate matter with next-day physical activity and sedentary behaviour in older adults: a micro-longitudinal study

Micro-longitudinal preprint: 67 UK adults aged 50-85 wore personal air-quality monitors (waking hours) and 24-h accelerometers for five days (304 person-days). Within-person mixed models: a higher-than-usual day (+1 IQR, i.e. +2.6 $\mu\text{g m}^{-3}$ PM_{2.5} above the person's mean) was followed by 8 min less light activity (95% CI -14 to -2) and 483 fewer steps (-855 to -82); for PM₁₀ (+2.8 $\mu\text{g m}^{-3}$), 8 min less LPA (-13 to -3), 8 min more sedentary time (0 to 15) and 412 fewer steps. No MVPA association, and none with area-level PM. Not peer reviewed; the monitor model and any calibration are not stated; a 2.6 $\mu\text{g m}^{-3}$ contrast is within typical optical-sensor error, so residual weather or illness confounding of day-to-day behaviour is the main threat. Relevance: high - the association exists only in personal-sensor exposure, not area estimates.

doi: 10.64898/2026.09.23.26363783

Chen et al. 2026 Environ Sci Technol Lett

Solid Waste Burning: An Overlooked Atmospheric Source of p-Phenylenediamine Antioxidants and Their Quinones

Field sampling of uncontrolled open burning and controlled incineration of solid waste with suspect screening annotated 47 PPD-related features in PM_{2.5} (43 in open-burning, 27 in incineration; 20 confirmed). Open burning emitted parent PPDs (median 11.54 $\mu\text{g/g}$ PM_{2.5}); incineration had relatively more quinones (median 2.50 $\mu\text{g/g}$). A screening-level national estimate attributes >99.7% of the 10 targeted PPD + PPD-Q emissions from waste burning in China to open burning. Emission factors rest on few events and the national scaling on activity data of uncertain quality. Relevance: moderate - 6PPD-quinone toxicity makes this a composition-specific hazard PM_{2.5} mass sensors cannot register; open burning is detectable as short plumes in dense networks.

doi: 10.1021/acs.estlett.6c00796

Guo et al. 2026 Atmosphere

Atmospheric Environment of the Northern Tianshan Urban Agglomeration, Xinjiang: A Comprehensive Review of Pollutant Characteristics, Photochemical Processes, and Health Implications

Narrative synthesis of air-quality research in the Northern Tianshan agglomeration, grading evidence as local, regional Central Asian, or extrapolated. Aerosol shifts from dust-dominated coarse mode in spring to secondary inorganic and carbonaceous fine mode in winter; BC is local fossil combustion, BrC sources are speculative; ozone regime diagnoses are thin and there are no radical or oxidative-potential measurements. Local epidemiology is essentially absent, so the health section rests on extrapolation. No systematic search. Relevance: moderate - an arid, dust-coal mixture region where optical sensors need dust-aware correction and where a dense network would fill a documented gap.

doi: 10.3390/atmos17100961

He B. et al. 2026 Atmos Environ

Contrasting seasonal behavior and inhalation risks of atmospheric PAHs, PAEs, and OPEs in a heating-influenced urban atmosphere

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes seasonal behaviour and inhalation risk of three semivolatile organic classes in a heating-season city. Composition data of this kind are what mass-only networks cannot supply. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122394

He et al. 2026 J Environ Manage

Interpretable machine learning for spatially differentiated nonlinear relationships between PM_{2.5} and landscape patterns in an urban-rural integrated zone

Random forest with SHAP relates PM_{2.5} to grey-green landscape metrics across an urban-rural zone of Xuchang. Three regimes emerge: elevation dominates in mountains; in dense cores the positive association with mean building height strengthens above 35-40 m; in transition zones green-space porosity of 0.18-0.22 (plains) or 0.12-0.15 (hills) and blue-space connectivity go with lower PM_{2.5}. The PM_{2.5} surface source and resolution are not given in the abstract, model skill is not reported, and SHAP thresholds on a single city are associations, not design standards - the authors say as much. Relevance: low-moderate; building-height and porosity thresholds are testable with a dense street-level sensor network rather than a gridded product.

doi: 10.1016/j.jenvman.2026.130981

Kuntz et al. 2026 J Hazard Mater

Dynamics of tire-related chemicals in roadside aerosol over one year in Leipzig, Germany-influencing factors, potential source materials and comparison with rural aerosol

119 roadside PM₁₀ samples over 14 months were analysed for 40 tire-related chemicals. Medians: 6-PPD 0.61 pg/m^3 , 6-PPD-quinone 2.9 pg/m^3 , diphenylguanidine 90 pg/m^3 , benzothiazolesulfonic acid 41 pg/m^3 . Concentrations spanned about four orders of magnitude, with precipitation the dominant control, modulated by temperature, sunshine, wind and oxidants. Tread particles carried one to two orders of magnitude more TRC per mass than roadside aerosol, and rural levels were up to tenfold lower. Traffic-normalised concentrations are offered for extrapolation, but one roadside and one rural site cannot support transfer across fleets and climates. Relevance: moderate - a toxic tyre-wear fraction that optical sensors cannot distinguish within PM₁₀.

doi: 10.1016/j.jhazmat.2026.143749

Liu et al. 2026a J Environ Sci (China)

Persistent cold air pools and associated sustained haze pollution in Taiyuan Basin

Builds a refined detector for persistent cold-air pools from multi-station and sounding data in the Taiyuan Basin (2018-2022), validates WRF for PCAP meteorology, and identifies 56 events, mostly September-March, winter ones commonly lasting >48 h. Events carry stronger surface inversions, larger valley heat deficit and suppressed ventilation; PM_{2.5} peaks lag heat-deficit intensification by 12 h ($R = 0.55$). Warm advection aloft maintains PCAPs and lee-side subsidence warming reinforces them; strong cold advection ends them. Limitation: $R = 0.55$ on a single basin, with emissions held implicit, so the 12-h lag is descriptive. Relevance: moderate - a 12-h lag from a sounding-derived metric is a usable forecasting feature, and basin networks must site nodes at more than one elevation to see the pool.

doi: 10.1016/j.jes.2026.03.027

Liu et al. 2026b J Environ Sci (China)

Metropolitan atmospheric microplastics: Characteristics and its inputs to fluvial system

Deposition sampling across urban, suburban and exurban Chengdu: 246, 192 and 177 items/m²/day respectively, dominated by fibres and fragments. PCA finds three source patterns (73.9% of variance), the largest a mixed daily-plastic-products source (55.1%). Abundance tracks precipitation intensity, weakly wind direction, not temperature or RH, and shows no relation to other air pollutants. Rainy-season input to the river system is estimated at 689–1,088 kg. Limitation: deposition flux, not airborne concentration or inhalable size fraction, so there is no exposure metric; identification method and lower size cut are not given in the abstract. Relevance: low - the absent correlation with regulated pollutants means PM networks are no proxy for airborne plastics.

doi: 10.1016/j.jes.2026.03.077

Luo et al. 2026 Environ Pollut

Primary nitrated phenol emissions from heavy-duty diesel vehicles based on real-world activity data in Hunan, China

Combines tandem filter/PUF exhaust emission factors with real-world heavy-duty diesel activity data to inventory primary nitrated phenols in Hunan for 2025: 1,196.8 kg/yr, 48.3% from container trucks, 22.9% freight trucks, 17.7% buses; China IV and V vehicles 84.1%. 4-Nitrophenol dominates (334.9 kg/yr). Monthly totals range 76.8 kg (Feb) to 114.4 kg (Dec), and five cities emit 53.9%. Limitation: emission factors from a limited test fleet are scaled to a province, the particle/gas split is not reported in the abstract, and no uncertainty is given. Relevance: low - nitrated phenols are a light-absorbing BrC component near freight corridors, relevant to multi-wavelength absorption channels, but the mass involved is trivial against PM_{2.5}.

doi: 10.1016/j.envpol.2026.129268

Ma M. et al. 2026 Atmos Environ

Atmospheric Redistribution of Microplastics Reshapes Environmental Risk Across China

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes atmospheric transport of microplastics and its effect on risk across China. Airborne microplastics are part of the coarse fraction optical sensors mis-size. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122397

Mao et al. 2026 Atmosphere

Seasonal and Pollution-Level Variations in PM_{2.5}-Bound PFASs and Associated Health Risks in Jinan, a Typical Northern Chinese Industrial City

163 PM_{2.5} filters over a year (Dec 2023–Nov 2024) in Jinan, 17 PFASs by LC-MS/MS. Total PFAS ranks winter 133.8 > autumn 123.6 ~ summer 123.5 > spring 97.1 pg/m³; in summer and autumn concentrations were numerically higher on clean than on slightly polluted days. No monotonic relation of total PFAS with single meteorological variables; compound responses depend on chain length. PFOA and short-chain PFCA make up 73.1% of estimated intake and PFOA 95.6% of risk. Limitation: 'numerically higher' differences are not tested, one site, and risk uses reference doses for ingestion-derived toxicity. Relevance: moderate - a PM_{2.5} hazard that is decoupled from PM_{2.5} mass is invisible to any mass sensor, whatever its calibration.

doi: 10.3390/atmos17100967

Mielnik et al. 2026 Research Square (preprint)

Comparing geospatial interpolation techniques for estimating ambient PM_{2.5} mass concentrations from sugarcane burning in South Florida

Low-cost sensors at 46 locations (8 Jun 2024 - 1 Dec 2025) spanning burn and non-burn seasons in the Everglades Agricultural Area; IDW and kriging variants compared by leave-one-out cross-validation for 24 h PM_{2.5}. Daily R² ranged 0–0.99; IDW won on most days, and a per-day best-model ensemble was most robust. Smoke presence raised estimated PM_{2.5} by 1.65 µg m⁻³ (SE 0.13), concentrated inland near burn areas; the coast was cleanest. Limitation: sensor type, calibration and humidity correction are not stated in the abstract - in smoke, optical sensor bias is composition-dependent, so interpolation error and sensor error are confounded; LOOCV on a network this sparse rewards smoothness; preprint. Relevance: high - a direct test of how far a rural LCS network can be stretched into a health-study exposure surface for an episodic agricultural source.

doi: 10.21203/rs.3.rs-10743669/v1

Momen et al. 2026 Research Square (preprint)

Explainable Machine Learning for Predicting Thermal Discomfort from Multipollutant Exposure on Dhaka Roadways

Preprint. Roadside PM_{2.5}, noise and microclimate were measured on Dhaka arterials in the monsoon alongside thermal sensation votes; mean PM_{2.5} was 58.35 µg m⁻³ and noise 82.96 dB, and 99.46% of votes were hot, warm or slightly warm. ML classifiers reached AUC 0.96 with SHAP attributing discomfort across heat, PM and noise. The outcome is thermal sensation, not a health endpoint; sample size, instrument type and validation scheme are not in the abstract, and an AUC of 0.96 on a 99% prevalent outcome says little. Not peer reviewed. Relevance: low; co-located PM, noise and heat is the multi-stressor sensor package pedestrian-exposure networks increasingly deploy.

doi: 10.21203/rs.3.rs-10487427/v1

Munteanu et al. 2026 Sensors (Basel)

Evaluating Graph-Centrality Context for Station-Level Correction of CAMS Air Quality Reanalysis in Lombardy from 2018 to 2021

Corrects CAMS European reanalysis to ARPA Lombardia station observations (2018–2021) by treating stations as nodes of a multiplex graph and feeding six centrality measures (including an OpenStreetMap-context adapted PageRank) into GTWR, XGBoost and random forest with spatio-temporal kriging, under station, temporal and combined holdout. Station-holdout XGBoost reaches R² up to 0.66 for NO₂; OSM-context centrality helps NO_x and PM_{2.5}, is mixed for NO₂, and helps PM₁₀ only temporally with RF-STK. Limitation: the abstract gives no PM_{2.5} R² or RMSE, gains are described qualitatively, and centrality is computed on the regulatory station layout itself, so transfer to a denser or differently sited network is untested. Relevance: moderate - graph features of node placement are directly computable for low-cost networks.

doi: 10.3390/s26185844

Na et al. 2026 Atmos Environ

Satellite-based estimation of PM_{2.5} chemical composition under sparse observations in Seoul metropolitan area

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes satellite-based estimation of PM_{2.5} chemical components from sparse speciation observations. Relevant to satellite-surface fusion and to composition-specific health models. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122402

Nassir et al. 2026 Environ Pollut*A multi-sensor satellite assessment of air quality during the 2026 Middle East Conflict across nine regional countries*

A dual-baseline design (14-month annual and seasonally matched prior year) applied to the 39-day 2026 conflict period across nine countries (~217 million people). Domain-mean anomalies: NO₂ -15.1%, SO₂ -45.6%, CO -8.6%, AOD -19.6%, CH₄ +0.37% (noise), ordered by atmospheric lifetime - the signature of an emission change; cold spots sit over Iranian, Kuwaiti and Qatari hydrocarbon fields. The NO₂ burden estimate (-7,849; 95% CI -35,641 to +13,551) crosses zero, and the PM_{2.5} burden is called indeterminate because dust and anthropogenic aerosol cannot be separated at CAMS resolution. Limitation: 39 days, no surface PM validation; a wind reversal is documented. Relevance: moderate - an honest statement that column AOD cannot apportion PM in dust-dominated regions without ground speciation.

doi: 10.1016/j.envpol.2026.129258

Niamien et al. 2026 Atmos Pollut Res*Spatio-temporal analysis of the physical properties of aerosols in urban and rural environments in Cote d'Ivoire*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes urban-rural aerosol physical properties in Cote d'Ivoire. West Africa is among the least monitored regions, where low-cost networks carry most of the observational load. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.apr.2026.103220

Oliveira et al. 2026 Atmos Pollut Res*Nanomineralogical and geochemical signatures of coal-derived particles in playground dust*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 1 Oct with no abstract; PubMed efetch, Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes nanomineralogy of coal-derived particles in playground dust. Relevant to child resuspension exposure, which ambient nodes do not measure. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.apr.2026.103209

Rawat et al. 2026 Atmos Environ*Size-resolved aerosols from residential fuel combustion in the northwestern Himalayas: potential effects on light absorption and human health*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes size-resolved aerosol from household fuel combustion and its absorption and health implications. Relevant to household-air-pollution exposure, today's tuberculosis burden paper being the health side of the same source. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122399

Shahadat et al. 2026 Environ Geochem Health*Spatiotemporal variability, trends, and health risk assessment of ambient PM_{2.5} and PM₁₀ in Bangladesh during 2018-2025.*

Analysis of 2018-2025 PM_{2.5} and PM₁₀ from the 16 Department of Environment continuous stations in Bangladesh, with seasonal Mann-Kendall trends and ELCR/HQ risk indices. Every station exceeded national standards and WHO guidelines every year. Network annual mean PM_{2.5} fell from 91.54 to 77.64 $\mu\text{g m}^{-3}$ (-15.2%) and PM₁₀ from 155.7 to 132.3 (-15.0%), non-monotonically, rebounding after the 2020 minimum; winter/monsoon ratios were ~4.2 (PM_{2.5}) and 3.0 (PM₁₀). Most seasonal trends were not significant. The authors note kriging performed poorly and was used only for display - 16 stations cannot map a country of 170 million. ELCR applied to PM_{2.5} mass is a screening construct, not an epidemiological risk. Relevance: the station-data quality control is undescribed, and this network is exactly what low-cost sensor densification in Bangladesh would calibrate against.

doi: 10.1007/s10653-026-03506-z

Sheidaei et al. 2026 Environ Sci Technol*A Multi-Stage Deep Learning Framework for Daily PM_{2.5} Estimation at 100 m Resolution Across the Contiguous United States, 2000-2024*

Three-stage pipeline: a reanalysis-informed U-Net fills missing satellite AOD, a bidirectional LSTM refines its temporal structure and terrain downscales it to 100 m; a multistream network then predicts daily PM_{2.5} over >766 million cells for 2000-2024. Site-held-out validation: R₂ = 0.82, RMSE = 2.85 $\mu\text{g m}^{-3}$, MAE = 1.84 $\mu\text{g m}^{-3}$; spatial R₂ 0.94, temporal R₂ 0.78. Limitation: the 100 m detail comes from downscaled AOD and land-use covariates, not from 100 m observations - validation sites are regulatory monitors, which cannot test sub-kilometre gradients near roads or smoke plumes; the temporal R₂ (0.78) is the binding figure for short-term epidemiology, and performance during wildfire days is not reported. Relevance: high - a 25-year 100 m baseline is the natural prior that dense low-cost networks should be fused against, and the place their residuals are informative.

doi: 10.1021/acs.est.6c05375

Shoukat et al. 2026 Atmos Pollut Res*Suspended Atmospheric Microplastics in the Outskirts of Seoul City: Seasonal Variability, Correlation with Meteorology*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 1 Oct with no abstract; PubMed efetch, Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes seasonal suspended (airborne) microplastic measurements near Seoul and their meteorological correlates. Complements the Chengdu deposition record in this issue with an airborne concentration metric. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.apr.2026.103222

Singh et al. 2026 Atmos Environ*Aerosol Absorption Coefficients Across Global Sites: Impacts of Chemical Composition, Emissions, and Meteorology*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 29 Sep with no abstract; PubMed efetch, Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates a multi-site analysis of aerosol absorption coefficients against composition, emissions and meteorology. Relevant to absorption-based black-carbon sensors and their site-dependent mass-absorption cross-sections. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122391

Surdu et al. 2026 Environ Sci Technol*Summertime Aerosol Composition in Southern Greenlandic Fjords: Impacts of Sources and New Particle Formation Events*

First aerosol composition measurements in the fjords near Narsaq, South Greenland, from two summer ship campaigns using nanoelectromechanical-systems FTIR. Median organic matter was $1.1 \mu\text{g m}^{-3}$, above High Arctic and mid-latitude coastal sites; PMF split the aerosol into primary marine (53%) and MSA-bearing secondary marine (47%) factors, amines exceeded open-ocean levels, and NPF events shifted composition toward secondary material. Campaign length and sample counts are not in the abstract, and two summers cannot separate interannual bloom timing from sampling differences. Relevance: marginal for health; NEMS-FTIR is a low-mass, high-time-resolution composition method that could pair with network optical sensors at supersites.

doi: 10.1021/acs.est.6c09104

Vukic et al. 2026 Environ Monit Assess*Integrated assessment of road and maritime traffic impacts on urban air quality and noise in a port-city environment*

Ten sites in Split, Apr-Sep 2024: 154 port and 66 road short-duration observations of $\text{PM}_{2.5}$, PM_{10} , CO, traffic counts and (road only) noise. Port means exceeded road means ($\text{PM}_{2.5}$ 14.77 vs 11.64; PM_{10} 18.25 vs 14.70 $\mu\text{g m}^{-3}$) but not significantly ($p = 0.22, 0.23$). Vessel count was weakly associated with PM_{10} ($R^2 = 0.082$) and stayed positively associated with port $\text{PM}_{2.5}/\text{PM}_{10}$ in cluster-robust models; wind speed lowered PM. Limitation: spot measurements with uncalibrated low-cost sensors, no reference co-location, and vessel count ignores engine type, berth time and fuel - the R^2 says ship numbers are a poor proxy for ship emissions. Relevance: moderate - the authors frame LCS correctly as screening-level; port sources need continuous nodes with plume-resolving time resolution, not spot visits.

doi: 10.1007/s10661-026-15986-8

Wang et al. 2026a J Environ Sci (China)*Organosulfate-driven amplification of secondary organic aerosol formation from isoprene oxidation intermediates: Coupled effects of SO_2 and relative humidity*

Smog-chamber methacrolein/OH photooxidation at high (61-65%), medium (20-22%) and low (9-12%) RH with varied SO_2 . Particle mass, peak number, geometric mean diameter and SOA yield all rise with initial SO_2 at every RH; with SO_2 present, SOA mass rises as RH falls, attributed to lower wall loss of hydrophilic species and less dilution of aerosol acidity. ATR-FTIR shows organosulfates and inorganic sulfate in the particle phase. Limitation: chamber concentrations and wall-loss corrections dominate yield estimates, and no yields are reported in the abstract; transfer to ambient isoprene- SO_2 mixing is assumed. Relevance: low-moderate - acid-catalysed SOA is part of why the optical-to-mass ratio of low-cost sensors drifts with season and RH.

doi: 10.1016/j.jes.2026.02.001

Wen and Chang 2026 Environ Pollut*Estimating marine $\text{PM}_{2.5}$ concentrations and exploring the drivers over the eastern China seas using multi-source data and machine learning*

BPNN, random forest and XGBoost compared under sample-, spatial- and temporal-held-out validation for $\text{PM}_{2.5}$ over the Bohai, Yellow and East China seas; XGBoost best. Estimated fields fall west-to-east and north-to-south with maxima in Bohai Bay and off Shandong, a declining trend with winter maxima, a visible 2020 drop, and SHAP attributing spring-autumn variance to AOD and winter variance to meteorology. Limitation: there are almost no marine ground monitors, so training and validation both lean on coastal land sites - held-out skill over open water is unverified and the 'marine' field is an extrapolation; no shipping-emission predictor. Relevance: moderate - the obvious use case for buoy- or vessel-mounted low-cost nodes is precisely this validation gap.

doi: 10.1016/j.envpol.2026.129276

Zheng et al. 2026 J Environ Sci (China)*Comparative analysis of multiple health risks from different air pollution events*

Online and offline measurements in Beijing (Feb-May 2019) classify haze (83% secondary), firework (31% firework) and dust (50% dust) events and compare risk metrics against clean days. Haze and firework particles deposit mainly tracheobronchially and pulmonary, dust in extrathoracic and TB regions; ultrafines are $\geq 81\%$ of deposited number in every event. Carcinogenic risk exceeds $1e-4$ in all events (7.6, 4.2 and 1.5 times clean-day $2.3e-4$); hazard index reaches 241, 100 and 37 (firework, dust, haze) vs 23 clean. $\text{PM}_{2.5}$ OP is 0.74-0.82 vs 0.47 nmol/min/m³ clean. Limitation: one spring, one site, few events per type; HI and TCR are deterministic screening indices with default toxicity values, not health outcomes. Relevance: moderate - event rankings flip by metric, which a mass-only network cannot reproduce.

doi: 10.1016/j.jes.2026.02.033

Cardiovascular & metabolic

8 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (8 OF 8 RECORDS IN THIS SUBTOPIC)

Cosenza et al. 2026 medRxiv (preprint)*Associations of Outdoor Air Pollution With Cause-specific Mortality in Metastatic Prostate Cancer: A cohort study*

33,384 men with metastatic prostate cancer (11 US registries, diagnosed 2000-2015, followed to 2018); 1-km $\text{PM}_{2.5}$ and NO_2 at the diagnosis address; Cox and IPW cause-specific models. Per IQR $\text{PM}_{2.5}$, cardiovascular mortality HR 1.04 (1.02-1.07), with 4th/5th vs 1st quintile HR 1.18 and 1.21; ADT-treated men with high vs low $\text{PM}_{2.5}$ had HR 1.19 (1.05-1.34), interaction $p = 0.051$. NO_2 was associated only with prostate-cancer death in the top quintile (HR 1.07, 1.01-1.13). Limitation: one address at diagnosis, no smoking or cardiovascular history in registry data, and cause-of-death coding in metastatic cancer is unreliable; the IQR is not stated in the abstract; preprint. Relevance: low-moderate - a high-risk population whose acute exposures personal sensors could resolve.

doi: 10.64898/2026.09.29.26364034

Jin et al. 2026 Neuroepidemiology*Ambient air pollution and stroke in two population-based cohorts: roles of systemic inflammation and polygenic risk.*

Two separate analyses. CHARLS 2015 cross-section ($n=21,095$): prevalent self-reported stroke OR 1.26 (95% CI 1.04-1.53) for $\text{PM}_{2.5}$ and 1.31 (1.08-1.59) for PM_{10} , increments not in the abstract, with positive multiplicative interaction with hs-CRP. UK Biobank ($n=368,644$): incident ischaemic stroke HR 1.29 (1.15-1.45) per $5 \mu\text{g m}^{-3}$ $\text{PM}_{2.5}$; high PRS plus high $\text{PM}_{2.5}$ HR 1.44 (1.32-1.57), without a formal interaction test. The UKB estimate is large for a $5 \mu\text{g m}^{-3}$ contrast in a low-exposure cohort and conflicts with Wang et al. today (null for stroke per IQR), which points to specification sensitivity. Relevance: moderate - inflammation-stratified susceptibility argues for better exposure resolution in high-CRP groups.

doi: 10.1159/med/adwag036

Kriti et al. 2026 Front Public Health*Ecological associations between ambient air pollution, gut microbiome composition, and metabolic markers in urban populations of central India.*

Cross-sectional ecological study of 95 adults from three Bhopal localities: station PM_{2.5}, PM₁₀, NO_x and SO₂ assigned by locality, T2D markers, and full-length 16S rRNA gut profiles. NO_x correlated positively with glycaemic indices; PM_{2.5} and PM₁₀ showed inconsistent, non-directional associations; community structure was broadly similar and most genus differences did not survive multiple-testing correction. With three exposure levels shared by all participants in a locality, the design cannot separate pollution from anything else that differs between neighbourhoods - the authors call it hypothesis-generating. Relevance: low; a textbook case where personal or dense sensor exposure would be the minimum to make the question answerable.

doi: 10.3389/fpubh.2026.1834285

Lin et al. 2026 Front Public Health*Short-term lagged exposure to ambient air pollutants and hospitalization costs among AMI patients: health-economic evidence from a subtropical city*

Single-centre retrospective cohort of 3,650 PCI-treated AMI admissions (2018-2024) in Jiangmen, where concentrations met China Grade II. Gamma GLMs with FDR correction: SO₂, PM_{2.5}, PM₁₀, NO₂ and CO were associated with higher hospitalisation cost; SO₂ largest (2.40%, ~729 CNY per log unit); PM_{2.5} and PM₁₀ peaked at lag 4-5; O₃ raised costs at lags 3 and 5 in 1,590 never-smokers. PM-specific percent changes and CIs are not in the abstract. Cost under DRG/DIP is a weak severity proxy, confounded by coding and length of stay, and city-wide monitors give no spatial contrast. Relevance: low; illustrates why lag structures need higher-resolution exposure.

doi: 10.3389/fpubh.2026.1939605

Rivera-Mariani et al. 2026 medRxiv (preprint)*Inflammatory Burden, Structural Exposure, and Racial Disparities in Incident Heart Failure: Evidence from the Multi-Ethnic Study of Atherosclerosis*

6,785 MESA participants followed ~16 years (485 HF events). IL-6, D-dimer, CRP and fibrinogen each predicted HF (HR 1.13-1.20 per SD); a composite index gave HR 1.25 (1.12-1.39); biomarker adjustment reduced the Black-White log-hazard gap by 57.5%. Associations were robust to PM_{2.5}/NO₂ co-adjustment in a geocoded subcohort. Limitation for this watch: air pollution is a sensitivity covariate only - no PM effect estimate is reported, and 'structural exposure' is inferred from a null interaction, not measured; single baseline biomarkers; preprint. Relevance: low - admitted because it tests whether PM explains part of an inflammatory pathway and reports that it does not change the estimates.

doi: 10.64898/2026.09.28.26364238

Sun et al. 2026 Indoor Air*Solid Fuel Use and Risks of All-Cause and Cardiovascular Disease Mortality in Urban Northern China: A 12-Year Cohort Study*

Retrospective cohort of 39,054 urban residents in four northern Chinese cities followed 1998-2009: 1,435 non-accidental and 677 cardiovascular deaths. Each additional year of solid-fuel heating was associated with HR 1.15 (95% CI 1.14-1.17) for non-accidental, 1.15 (1.13-1.18) for cardiovascular and 1.16 (1.12-1.20) for cerebrovascular mortality; cooking HRs were slightly higher; adults ≥60 were most sensitive. A 15% hazard per year of use is implausibly steep for a cumulative-exposure metric and points to confounding by socioeconomic position, housing age and co-exposure; fuel-years is self-reported and no indoor PM was measured. Relevance: moderate - the exposure contrast is real, but quantifying it needs the indoor sensors this study lacks.

doi: 10.1155/ina/9645092

Wang et al. 2026 Front Public Health*Long-term PM_{2.5} exposure and incident heart failure in the UK Biobank: relative and absolute risk estimates.*

149,410 UK Biobank participants, median 17.6 y follow-up, 4,964 incident heart-failure and 3,795 ischaemic-stroke events. PM_{2.5} per IQR: HR 1.045 (95% CI 1.005-1.086) for heart failure; top vs bottom quartile HR 1.133 (1.037-1.238), approximately linear. No effect modification by diabetes, CKD, obesity, multimorbidity or CRP, and no pollutant was associated with ischaemic stroke. The 20-year absolute risk difference between extreme quartiles was ~0.2 per 100. Exposure is a single-year LUR surface over a narrow UK range, and the null stroke result contrasts with Jin et al. today in the same cohort. Relevance: moderate - reporting absolute risk alongside HR is the right norm.

doi: 10.3389/fpubh.2026.1864868

Yu et al. 2026 Public Health*Effect modification of greenness on the associations between air pollutants and out-of-hospital cardiac arrest: Evidence from a multi-city study*

Individual-level time-stratified case-crossover of 62,915 OHCA cases in Chinese cities, 2020-2023, with exposure at onset location. Pollutants were associated with OHCA and greenness attenuated the association: per 0.1 NDVI(250 m), the percent change in OHCA odds per 10 µg m⁻³ fell by 1.62% (95% CI 0.39-2.84) for PM_{2.5} and 1.13% (0.24-2.01) for PM₁₀; results held at 500 and 1000 m and across lags. The main-effect estimates are not in the abstract, exposure is area-level at onset rather than personal, and greenness is confounded with SES and baseline pollution. Relevance: moderate - greenness-dependent slopes argue that sensor-derived exposure contrasts should be stratified by vegetation context.

doi: 10.1016/j.puhe.2026.106515

Neuro / mental health**6 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (6 OF 6 RECORDS IN THIS SUBTOPIC)

Bloomberg et al. 2026 medRxiv (preprint)*Daily Exceedance of World Health Organization Air Pollutant Guidelines and Cognitive Aging*

English Longitudinal Study of Ageing, 12,235 adults 50+ (waves 1-10, 2002-2023). Exposure is the number of days above WHO daily guideline values in the year before baseline, modelled at the residence, adjusted for the annual mean. NO₂ exceedance days (75th percentile vs median) predicted faster 14-year decline: memory -0.09 SD (-0.17 to -0.01), fluency -0.10 SD (-0.19 to 0.00). PM_{2.5} and PM₁₀ exceedance days were not associated with decline or baseline cognition. Limitation: exceedance count and annual mean are highly collinear, so the 'peak beyond mean' effect is fragile; one exposure year; attrition in ageing cohorts is cognition-dependent; preprint. Relevance: moderate - a null for PM peak frequency is exactly what high-time-resolution networks are positioned to retest.

doi: 10.64898/2026.09.23.26363787

Han et al. 2026 Int J Epidemiol*Air pollution and criminal behavior: a nationwide case-crossover analysis in China*

Time-stratified case-crossover of 483,718 crime events in China (2013-2019) with conditional logistic regression per pollutant and weighted-quantile-sum for the mixture. Per IQR on the event day, crime rose 1.66% for PM_{2.5}, 1.88% PM₁₀, 4.03% NO₂, 2.03% CO, 2.81% SO₂ and 1.30% O₃; WQS index +3.22% per unit. Associations were stronger in the cold season, the north and lower-SES regions, and for violent crime and theft. Limitation: every pollutant is 'positive', which is what shared confounding by weather, daylight and outdoor activity patterns looks like; exposure is assigned at a coarse scale to events of unknown offender residence; no CIs in the abstract. Relevance: low - a behavioural endpoint with acute timing, where personal or hyperlocal exposure would separate PM from co-pollutants.

doi: 10.1093/ije/dyag214

Januel et al. 2026 J Neurol Neurosurg Psychiatry

Air pollution and the risk of multiple sclerosis relapses in the Paris region, the CONFISEP study

1,152 adults with relapsing MS (898 relapses, 2018-2021) from the French MS registry; weekly Airparif street-level exposure at lags 0-15 weeks, analysed by self-controlled case series and cohort models adjusted for time-varying DMT, infections, temperature, sunlight and lockdown. PM₁₀ raised relapse risk at the 2-week (IRR 1.28, 95% CI 1.05-1.57) and 11-week lags (1.26, 1.02-1.54); PM_{2.5} at 2 weeks (1.20, 1.03-1.39); NO₂ at 2, 3 and 12 weeks. The exposure increment is not stated in the abstract, and with 16 lags x 4 pollutants the isolated 11-12-week hits are likely multiplicity, as the authors concede. DMTs dominated. Relevance: moderate - week-scale personal or neighbourhood sensing would sharpen exactly this lag structure.

doi: 10.1136/jnnp-2026-339534

Kwon 2026 Front Public Health

Fine particulate matter (PM_{2.5}) and suicide in South Korea: an instrumental variable analysis using thermal inversion.

Instrumental-variable panel analysis of South Korean suicide mortality 2015-2019 using thermal inversions as the instrument for PM_{2.5}. Each 1 µg m⁻³ raised suicide deaths by 0.66 per million (95% CI 0.10-1.22), about 2.94% of the rate; effects were larger in males, older adults, the less educated and non-metropolitan residents. The IV design addresses confounding by economic activity, but inversions also trap NO₂ and other pollutants and co-vary with cold, stagnant weather that may affect mood, so the exclusion restriction is questionable; temporal and spatial units are not stated in the abstract. Relevance: moderate - quasi-experimental PM designs need the exposure resolution that inversion-sensitive vertical sensing could supply.

doi: 10.3389/fpubh.2026.1909365

Shupler et al. 2026 Environ Sci Technol

Preconception, Prenatal and Postnatal PM_{2.5} Exposure and Neurodevelopmental Disorder Risk among Medicaid Recipients

1,547,244 Medicaid mother-child pairs (2001-2013), ZIP-code PM_{2.5} with distributed-lag models over preconception, pregnancy and the first three years. Per 10 µg m⁻³ cumulative prenatal PM_{2.5}, ASD HR 1.17 (95% CI 1.00-1.37), with a critical window at gestational weeks 23-31 (HR 1.09, 1.02-1.17). Prenatal exposure raised intellectual-disability risk in Hispanic children only (HR 1.57, 1.02-2.43). Postnatal ADHD (1.17, 0.94-1.46) and DCD (1.72, 0.97-3.07) estimates include the null; no other NDD was associated. ZIP-code exposure, claims-based diagnoses and residual area SES confounding limit the inference; the ASD lower bound sits at 1.00. Relevance: moderate - a third-trimester window argues for time-resolved residential exposure rather than annual means.

doi: 10.1021/acs.est.6c03766

Wang et al. 2026 Environ Int

Environmental exposures and mental health: longitudinal results from the German National Cohort (NAKO)

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 27 Sep with no abstract; Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates longitudinal analyses of environmental exposures and mental health in the German National Cohort; which exposures (air pollution, noise, greenness, heat) are included cannot be stated. NAKO's size (~200,000) makes it a priority retry. Nothing about exposure metric, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title.

doi: 10.1016/j.envint.2026.110539

Respiratory & allergic

5 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (5 OF 5 RECORDS IN THIS SUBTOPIC)

Carroll et al. 2026 Ann Am Thorac Soc

Global Tuberculosis Burden Attributable to Household and Outdoor Air Pollution

140 studies (to Aug 2024) pooled by random effects and Burden of Proof meta-regression. Household solid fuel: tuberculosis incidence RR 1.82 (95% CI 1.52-2.17), 2.25 in children and 2.62 in women. Short-term outdoor PM_{2.5}: RR 1.03 (1.00-1.05) per 10 µg m⁻³. Attributed to PM_{2.5} in 2023: 4.15 million cases, 23.6 million DALYs and ~535,000 deaths - 49.9% of global TB. The attributable fraction is driven by the household-fuel RR, which comes from fuel-type contrasts confounded by poverty, crowding and undernutrition, and is far larger than the outdoor estimate whose lower bound touches 1. Treat the 50% as an upper bound. Relevance: moderate - household PM measurement is the missing exposure link.

doi: 10.1093/annalsats/aaog299

Kulkarni et al. 2026 JAMA Netw Open

Short-Term Exposure to PM_{2.5} and Emergency Department Visits for Asthma and Chronic Obstructive Pulmonary Disease.

METADATA ONLY. Metadata-only record: PubMed carries only the one-line plain-language summary - a cohort study of short-term PM_{2.5} and asthma- and COPD-related emergency department visits in Singapore (Duke-NUS / SGH authors). No estimate, lag, exposure source or sample size is deposited; Crossref and Europe PMC returned nothing and Scholar Gateway failed with an identity error. This watch does not infer results from a title. Priority retry: Singapore's transboundary-haze PM_{2.5} is a clean natural experiment, and a tropical-humidity city is where sensor-derived exposure would be most biased.

doi: 10.1001/jamanetworkopen.2026.36306

Maunakea et al. 2026 Proc Natl Acad Sci U S A

Climate-amplified wildland-urban interface disasters create persistent internal metal mixtures associated with pulmonary impairment.

1,400 adults enrolled 6-18 months after the August 2023 Maui wildland-urban interface fires had 24 urinary metals and spirometry at the same visit. Burdens exceeded US reference levels - antimony 40-fold, manganese 3.8, barium 2.3, arsenic 2.2 - highest in the most affected areas. In WQS models each quantile of the positive mixture index (driven by As, Cd, Cu) raised the odds of FVC, FEV1, FEV1/FVC and FEF25-75 below LLN (OR 1.71, 1.53, 1.83, 1.84; CIs not in the abstract). Cross-sectional, no pre-fire baseline, urinary metals partly dietary (As) and smoking-related (Cd), and no air measurement links the burden to smoke or ash. Relevance: air PM mass would not have captured this; WUI smoke needs metal speciation, not just optical PM_{2.5}.

doi: 10.1073/pnas.2613208123

Shi et al. 2026 Indoor Air*Hidden Hazards at Home: Indoor Air Pollution and Adolescent Asthma - A Systematic Review and Meta-Analysis*

Meta-analysis of 77 observational studies of 11 indoor exposures and adolescent asthma. Dust mite OR 2.03 (95% CI 1.41-2.65; moderate certainty), pollen 1.70, mould/dampness 1.13 and pets 1.08. Cooking/heating was borderline (OR 1.09, 1.00-1.17; very low certainty) and open-fire sources raised risk (OR 1.26, 1.07-1.44). No association for climate-control appliances, VOCs or plants. Almost none of the combustion studies measured PM, so fuel type stands in for exposure, and GRADE certainty is mostly low. Symmetric CIs on OR scale suggest pooling on the linear scale. Relevance: moderate - the combustion signal is the one indoor PM sensors can quantify, and the evidence base lacks measurement.

doi: 10.1155/ina/7971301

Zouma et al. 2026 Toxicol Rep*Pulmonary function assessment of a community exposed to air pollution in an industrial area in Ouagadougou, Burkina Faso*

146 industrial-area residents vs 146 controls (Nioko 2), Nov-Dec 2023, with ATS/ERS spirometry. Symptom prevalence was higher in the industrial area (lower airway 89% vs 64%); pre-bronchodilator FEV1 %pred and FET% were lower ($p = 0.002$); restrictive defects were more common (OR 1.70, $p = 0.035$, no CI given), especially in women. Limitation: exposure is residence only - no PM or gas measurement, unadjusted univariate OR, and occupation, biomass cooking and socioeconomic differences between districts are uncontrolled. Relevance: moderate - West African industrial districts have essentially no PM monitoring; this is the kind of study a few calibrated nodes would turn from a residence contrast into an exposure-response.

doi: 10.1016/j.toxrep.2026.102344

Reproductive & developmental

8 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (8 OF 8 RECORDS IN THIS SUBTOPIC)

Bae et al. 2026 J Plast Reconstr Aesthet Surg*Exposure to ambient air pollutants during early gestation and phenotypic variation in non-syndromic microtia.*

Single-centre case-only study of 589 patients (625 ears) with non-syndromic microtia, relating spatiotemporally modelled exposure in gestational weeks 5-9 to bilateral involvement and anatomical severity. O₃ was nominally associated with bilateral involvement; PM₁₀, SO₂ and CO showed nominal inverse associations; after FDR only an inverse SO₂ - left-ear severity association in males survived, plus an SO₂ ear-side interaction. No PM association survives. Address at clinical registration (often years after gestation) misclassifies embryonic exposure, and without unaffected controls nothing can be said about occurrence. Relevance: low; a multiplicity-driven null for PM.

doi: 10.1016/j.bjps.2026.09.003

Gielen et al. 2026 Neonatology*The association of neighborhood socioeconomic status with birth weight is not confounded by ambient air pollution in twin pregnancies*

Population-based twin cohort stratified by placentation. In dichorionic twins, each EUR 3,000 higher neighbourhood median income raised birth weight by 46.83 g (21.92 g after adjusting for gestational age) and gestation by ~26 h; adding PM₁₀ and NO₂ did not change these estimates. PM effects themselves are not reported in the abstract, n is not stated, and kriged exposure over 1.55 km² sectors has limited contrast in Flanders. The finding concerns confounding of SES by PM, not a PM effect. Relevance: low; supports treating SES and PM as separable in birth-outcome models.

doi: 10.1159/neo/adsag016

Ha et al. 2026 Kidney Res Clin Pract*Developmental PM_{2.5} exposure and stage-dependent kidney senescence*

METADATA ONLY. Metadata-only record: PubMed (Korean Society of Nephrology) deposited title, authors and DOI on 1 Oct with no abstract; PubMed efetch, Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes kidney senescence after developmental PM_{2.5} exposure by life stage. Neither the species nor the exposure route is recoverable. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.23876/j.krcp.26.414

Kukshal et al. 2026 Indian J Pediatr*Maternal Exposure to Air Pollution During Early Pregnancy and Congenital Heart Disease: An Exploratory Case-Control Analysis from Delhi NCR*

1,556 participants (1,115 CHD cases, 441 controls) with residential exposure at conception and one month either side; multinomial regression, Bonferroni-corrected. Post-conception PM₁₀ was associated with all CHD (AOR 3.36, 2.05-5.52) and VSD (AOR 4.35, 2.41-7.83); preconception PM₁₀ with simple CHD (2.82, 1.69-4.71); 'poor AQI' gave AORs of 2.5-4.0. Limitation: the PM₁₀ contrast (per unit or category) is not given in the abstract; cases outnumber controls 2.5:1, implying hospital controls; Delhi-NCR monitor density makes residence-level contrasts small relative to city-wide covariance; ORs of 3-4 for an environmental exposure are far above the literature and suggest selection or confounding by SES. Relevance: moderate - a case for dense sensing where between-residence contrasts are the identifying variation.

doi: 10.1007/s12098-026-06363-x

Ma et al. 2026 J Environ Manage*Maternal PM_{2.5} exposure induces preeclampsia-like phenotypes in mice through placental angiogenic dysfunction and lncRNA-RUNX1 dysregulation*

C57BL/6J dams given intranasal heating-season PM_{2.5} through gestation developed raised blood pressure at E13.5 and E17.5, proteinuria, lower VEGF and PlGF, higher TNF- α , IL-6 and sFlt-1, and abnormal labyrinth architecture with impaired angiogenesis; ET-1 and NO were unchanged. Transcriptomics plus ChIP/RIP implicated lncRNAs NONMMUT087808.1 and NONMMUT044449.2 acting through RUNX1. Limitation: 3 mg/kg bolus intranasal instillation is far above any ambient-equivalent deposited dose and bypasses the nasal-pharyngeal filter, so the phenotype demonstrates sufficiency, not ambient relevance; one PM source, no filtered-air-plus-vehicle dose-response. Relevance: moderate - the sFlt-1/PlGF axis is measurable in human pregnancy and is the natural mediator to test against measured exposure.

doi: 10.1016/j.jenvman.2026.131066

Song et al. 2026 Environ Res*Role of Neighborhood Vulnerability and Opportunity in the Association Between Prenatal Air Pollution and Autism-Related Traits*

8,598 mother-child pairs from 35 ECHO sites; pregnancy and trimester PM_{2.5}, NO₂ and O₃ at residence; Social Responsiveness Scale analysed by GEE, stratified by census-tract SVI and Child Opportunity Index quintiles, with interaction $p < 0.10$ as 'suggestive'. O₃ interacted with SVI ($p = 0.07$) and COI ($p = 0.01$): beta 0.05 (-0.005 to 0.10) in lowest SVI and 0.04 (0.00 to 0.08) in highest COI. Trimester PM_{2.5} also showed interactions, but no stratum estimate was significant. Limitation: a liberal interaction threshold across many strata and trimesters invites false positives, and the O₃ signal appears in the advantaged strata, contrary to the hypothesis. Relevance: low - the PM_{2.5} finding is null within strata.

doi: 10.1016/j.envres.2026.125777

Wang et al. 2026 J Heart Lung Transplant*Long-term Exposure to PM_{2.5} and Risk of Adverse Pregnancy Events in Women with Pulmonary Hypertension: A Nationwide Multicenter Study*

751 pregnant women with pulmonary hypertension at 24 Chinese tertiary hospitals (2016-2022); 26.4% had maternal death or heart failure and 59.8% adverse fetal or neonatal events. Median 3-year PM_{2.5} 32.8 $\mu\text{g m}^{-3}$. Per 10 $\mu\text{g m}^{-3}$, OR 1.169 (95% CI 1.014-1.349) for maternal and 1.169 (1.018-1.348) for fetal/neonatal events, robust in two-pollutant models, strongest in late pregnancy. Retrospective and hospital-based, so referral patterns track region and pollution; the outcome is common, so ORs overstate relative risk; PH subtype and severity adjustment are not described. Relevance: moderate - a small, high-risk group where targeted indoor filtration with sensor feedback is a plausible intervention.

doi: 10.1016/j.healun.2026.09.021

Zhang et al. 2026 Environ Pollut*Second-trimester PM_{2.5} and PM₁₀ exposure is associated with neonatal metabolome alterations and elevated risk of pregnancy loss*

Propensity-matched cohort of 30,654 singletons (106 losses). Second-trimester PM_{2.5} OR 2.32 (95% CI 1.12-4.78) and PM₁₀ OR 1.86 (1.18-2.95) for pregnancy loss; increments not in the abstract. A five-metabolite risk score tracked second-trimester PM_{2.5} in single-window models ($b=0.052$) but not after mutual adjustment across windows ($b=0.003$, $p=0.29$); glycine, proline, valine and tyrosine were lower. With 106 events the CIs are wide, and the authors note single-window and sensitivity analyses were inconsistent. Neonatal metabolites cannot measure a lost pregnancy, so the mechanistic link is indirect. Relevance: low-moderate; window-specific estimates need residential exposure series, not annual surfaces.

doi: 10.1016/j.envpol.2026.129218

Mechanistic toxicology**13 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (13 OF 13 RECORDS IN THIS SUBTOPIC)

Cai et al. 2026 Ecotoxicol Environ Saf*Prenatal PM_{2.5} exposure disrupts cortical excitatory/inhibitory balance in male offspring by suppressing Lhx8-regulated cholinergic homeostasis and cell adhesion*

Male offspring of PM_{2.5}-exposed dams showed a persistent cortical shift toward inhibition (gephyrin up at PND 1, 7 and 21). RNA-seq and microarray identified 14 differentially expressed transcription factors; Lhx8 was consistently suppressed, with downstream ChAT loss, compensatory GABRA1 up-regulation and repression of adhesion genes (Cadm2, Cdh10, Pgrmc1, Cd36). Dose, route, PM source and litter-level analysis are not given in the abstract, males only, and no behavioural phenotype or Lhx8 rescue is reported, so causality within the proposed pathway is not tested. Relevance: low for sensing; supports the prenatal neurodevelopmental window seen in cohort data.

doi: 10.1016/j.ecoenv.2026.120862

Driencourt et al. 2026 Rev Mal Respir*The impact of post-exposure Streptococcus pneumoniae infections on pulmonary immunity: The role of air pollution and IL-20 cytokines in respiratory infections.*

French-language short review framing the experimental programme on how PM_{2.5} alters pulmonary immunity to Streptococcus pneumoniae, with IL-20-family cytokines as the candidate mediator. The abstract restates burden figures (>9 million deaths globally, 48,000 in France) and oxidative-stress/inflammation mechanisms, and reports no data. Relevance: low; flags IL-20 cytokines as a mechanistic target for PM-infection susceptibility studies.

doi: 10.1016/j.rmr.2026.07.009

Ferreira Vicente et al. 2026 Expert Rev Respir Med*Pulmonary black carbon accumulation: mechanisms, health impacts, and environmental sources*

Narrative review (searches to June 2026) of BC sources, physicochemistry, exposure assessment, biodistribution and tissue retention, with emphasis on tissue-based biomonitoring and label-free optical detection, and of respiratory, cardiovascular, neurological, developmental, metabolic and immune associations. The argument: tissue BC load could bridge external exposure estimates and biologically effective dose. Limitation: narrative, no quantitative synthesis or risk-of-bias assessment; tissue BC integrates an unknown exposure history, so it is a cumulative dose marker, not a validation of any particular exposure model. Relevance: low-moderate - internal BC dose is a possible ground truth for personal aethalometer and network-derived BC exposure.

doi: 10.1080/17476348.2026.2740205

Jiang et al. 2026 Redox Biol*Identification of shared and divergent molecular pathways linking ambient air pollutants to mortality and major chronic diseases through proteomic profiling*

Olink plasma proteomics in 48,645 UK Biobank participants: proteins shared across PM_{2.5}, PM₁₀, NO₂ and NO_x were enriched in immune regulation; PM_{2.5}-specific proteins in vascular dysfunction, PM₁₀ in DNA damage, NO₂ in apoptosis, NO_x in matrix remodelling. Mediation and MR flag proteins causally linked to outcomes (vascular remodelling for CVD, oxidative stress for CKD, cell adhesion for dementia, AGER for chronic respiratory disease and lung cancer). No effect sizes or mediated proportions in the abstract. MR establishes protein-to-disease causality, not pollutant-to-protein, and the four pollutants come from the same 2010-era LUR surfaces, so 'pollutant-specific' pathways may be collinearity artefacts. Relevance: low-moderate for sensing.

doi: 10.1016/j.redox.2026.104411

Jin et al. 2026 Front Immunol*ANKRD22 contributes to NF- κ B-associated inflammatory responses in alveolar type II epithelial cells during PM_{2.5}-triggered airway inflammation*

PM_{2.5} raised ANKRD22 in alveolar type II cells in a mouse airway-inflammation model and in A549 cells. AT-II-specific ANKRD22 knockout mice had less BALF leukocytosis and neutrophilia, histological inflammation and IL-6/CXCL1; siRNA knockdown in A549 lowered IL6, IL8, IL1B, CXCL1 and CXCL2. PM_{2.5} increased p50, phospho-IKK β , IkB α and RELA, and the non-canonical p52/RELB arm, all attenuated by ANKRD22 loss. Limitation: particle source, dose and dosimetry are not given in the abstract, A549 is a poor AT-II surrogate, and the knockout readouts are acute. Relevance: low - mechanism only; no exposure metric that a monitoring network could map onto.

doi: 10.3389/fimmu.2026.1894350

Kamyabi et al. 2026 J Dairy Sci

A novel controlled wood smoke exposure model to evaluate physiological responses to smoke-derived air pollution in dairy calves.

24 pre-weaned Holstein bull calves were randomised to 6 h in purpose-built enclosures with wood smoke ($\text{PM}_{2.5}$ 238.44 $\pm$ 62.97 $\mu\text{g m}^{-3}$) or clean air (9.94 $\pm$ 4.50), with CO_2 , CO, temperature and RH similar. Smoke-exposed calves had lower respiratory rate (17.50 vs 21.52 breaths/min) and higher tidal volume (0.90 vs 0.75 L), unchanged minute ventilation; BALF showed more granulocytes, CD68+ macrophages and NK cells, and circulating neutrophils fell. A single 6-h exposure with necropsy immediately after cannot address recovery or repeated smoke days; fresh wood smoke differs from aged wildfire plumes. Relevance: the enclosure $\text{PM}_{2.5}$ record gives a controlled dose, rare in agricultural wildfire work, and the design transfers to sensor-guided barn ventilation trials.

doi: 10.3168/jds.2026-29100

Kim et al. 2026 J Pharmacopuncture

Acorus gramineus Solander Preserves Corneal Epithelial Barrier Integrity against $\text{PM}_{2.5}$ -Induced EMT-Like Remodeling

Human corneal epithelial cells grown as 3D spheroids were exposed to $\text{PM}_{2.5}$ with or without an Acorus gramineus root extract. $\text{PM}_{2.5}$ destabilised spheroids without marked cytotoxicity, lowered transepithelial resistance, disrupted ZO-1, ZO-3 and occludin, induced EMT-like markers (N-cadherin, vimentin, MMPs) and cytokines, with TGF- β /Smad activation; the extract reversed these. Limitation: particle source and concentration are not stated in the abstract, no in vivo or tear-film model, and the therapeutic arm is a crude herbal extract. Relevance: low - the ocular surface is an underexamined direct-contact target, but nothing here quantifies dose.

doi: 10.3831/kpi.2026.29.3.314

Lee et al. 2026b Cell Biol Int

Curcumin-Loaded Nanospheres Attenuate Fine Dust (PM_{10})-Induced Cellular Toxicity by Suppressing Reactive Oxygen Species-Dependent Signaling in Human Keratinocytes

PM_{10} reduced HaCaT viability and raised ROS, ERK, p38 and NF- κB activity, shifted Bax/Bcl-2, cleaved caspase-3 and processed IL-1 β ; curcumin-loaded nanospheres at 1e-4 $\mu\text{g/mL}$ restored viability and blunted each step, as did NAC and pathway inhibitors. Limitation: this is a formulation paper - the PM_{10} arm is a generic oxidative-stress stimulus with no composition, dose-response or endotoxin control, and an effect at 1e-4 $\mu\text{g/mL}$ with none at higher unformulated doses is a non-monotonic claim that needs replication before it means anything. Relevance: low - logged for completeness of the dermal-endpoint strand, not as mechanism.

doi: 10.1002/cbin.70222

Liu et al. 2026c Environ Sci Technol

Bulk $\text{PM}_{2.5}$ Masks Oxidative Potential and Reactive Oxygen Species Responses by Metal-Organic Complexation in PM_1 and $\text{PM}_{1-2.5}$ Mixtures

Roadside PM_1 and $\text{PM}_{1-2.5}$ were tested separately and mixed. Mixing is antagonistic on every endpoint - interaction factor 0.65-0.81 for ascorbate OP, 0.80-0.93 for DTT OP, 0.10-0.88 for cellular ROS and 0.43-0.78 for mitochondrial superoxide - whereas metal-free biomass-burning PM is strictly additive. Cu $^{2+}$ -humic acid mixtures reproduce the antagonism (IF 0.22-0.95), and EPR shows complexed Cu forming nonlinearly with the HA:Cu ratio, explaining why studies disagree. Limitation: one roadside site, extract-based assays, and Cu-HA as the single model complex; whether in vivo lining fluid reproduces the complexation is untested. Relevance: high - bulk- $\text{PM}_{2.5}$ OP, the metric proposed as a toxicity-weighted complement to mass, understates the oxidative burden of its size fractions; size-resolved sampling is needed to interpret it.

doi: 10.1021/acs.est.6c09837

Liu et al. 2026d Environ Sci Technol

Metabolite-Associated Inflammation during Combined Exposure to Cooking Oil Fumes and Atopic Dermatitis: A Multi-Organ Spatial Metabolomics Study

A mouse model of atopic dermatitis with cooking-oil-fume co-exposure, profiled by untargeted and spatial metabolomics with in situ immunofluorescence across lung, blood, heart and kidney. AD+COF gave the largest rises in IL-6 and TNF- α and the largest metabolite shifts: succinate accumulation, xanthine and uric-acid depletion, glucose perturbation and spatial lipid remodelling, correlated across organs. Limitation, stated by the authors: no tracing, receptor blockade or mitochondrial assays, so inter-organ signalling is associative, and the high-concentration regimen limits environmental extrapolation; COF particle size and mass are not given in the abstract. Relevance: low - indoor cooking PM is a large exposure class, but this design cannot inform a dose metric.

doi: 10.1021/acs.est.6c11745

Onoda et al. 2026 Part Fibre Toxicol

Gene-environment interaction in neurodevelopment: Setd5 haploinsufficiency and prenatal carbon nanoparticle exposure synergistically induce ASD-like phenotypes

Pregnant C57BL/6N dams received intranasal carbon black nanoparticles (95 $\mu\text{g/kg}$) or water on GD5 and GD9; offspring were genotyped wild-type or Setd5 $^{+/-}$ and analysed by sex. Only male Setd5 $^{+/-}$ offspring exposed to CBNP showed a higher proportion with ASD-like behaviour (reduced social-novelty exploration, fixation on a familiar mouse) with selective loss of parvalbumin interneurons; either factor alone had minimal effect. Limitation: a single dose by intranasal bolus, a model particle without the metals or organics of ambient PM, and 'proportion of individuals' outcomes in small groups; group sizes are not in the abstract. Relevance: low-moderate - a gene-by-exposure synergy implies population risk is concentrated in susceptible genotypes that a mean exposure-response hides.

doi: 10.1186/s12989-026-00705-5

Saveleva et al. 2026 Mol Neurobiol

Urban $\text{PM}_{0.2}$ Affects Astrocytes by Inducing Xenobiotic and Oxidative Stress Responses

Primary astrocytes exposed to collected urban $\text{PM}_{0.2}$ at 50 $\mu\text{g/mL}$ for 24 h: 2,217 mRNAs altered, with strong induction of xenobiotic metabolism (CYP genes, Ahrr), Nrf2 oxidative stress responses and glutathione detoxification (GST superfamily, Nqo1); 354 proteins nominally changed, 142 concordant with mRNA. Reanalysis of published data shows microglia go pro-inflammatory while astrocytes go detoxification/antioxidant. Limitation: one dose, one time point, 50 $\mu\text{g/mL}$ is orders above any plausible brain dose, no particle-free extract control for organics vs core; proteomics is labelled exploratory. Relevance: low-moderate - reinforces that ultrafine/quasi-ultrafine number, not $\text{PM}_{2.5}$ mass, is the metric for neuro endpoints, which most LCS do not resolve.

doi: 10.1007/s12035-026-06232-w

Zhang et al. 2026c Free Radic Biol Med

$\text{PM}_{2.5}$ exposure impairs BMSC function and dysregulates mitochondrial oxidative stress: multi-omics insights into potential osteoporotic mechanisms

Rat BMSCs exposed to $\text{PM}_{2.5}$ lost viability, cytoskeletal integrity, proliferation and migration, with suppressed osteogenesis via β -catenin/Runx2, raised ROS and disturbed mitochondrial homeostasis. Transcriptomics gave 2,372 DEGs (911 up / 1,461 down) and metabolomics 171 DEMs, converging on cAMP signalling; forskolin rescued viability, ROS, membrane potential, migration and differentiation. Limitation: a single pooled $\text{PM}_{2.5}$ sample of unstated composition at unstated dose, applied directly to a cell type that in vivo sits behind the alveolar barrier - the rescue is strong internal evidence for the pathway but says nothing about delivered dose at the marrow. Relevance: low for sensing, moderate for the bone-outcome literature that epidemiology has begun to report.

doi: 10.1016/j.freeradbiomed.2026.10.004

Occupational & indoor

14 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (14 OF 14 RECORDS IN THIS SUBTOPIC)

Aljabri et al. 2026 Indoor Air*Indoor Air Quality and Exposure Characterization in Operational Waterpipe Cafes: Field Measurements of Particulate and Volatile Pollutants in Al-Madinah Al-Munawwarah*

Direct measurement in five operating waterpipe cafes on weekdays and weekends: mean weekday $PM_{2.5}$ $285.5 \pm 149 \mu g m^{-3}$ and PM_{10} $301.6 \pm 156 \mu g m^{-3}$, weekend TVOC $0.50 mg/m^3$, formaldehyde below short-term reference values. Variability tracked smoking intensity and ventilation. The $PM_{2.5}/PM_{10}$ ratio near 0.95 is the signature of a pure combustion source with no coarse fraction. Limitation: instrument make, calibration and whether an optical sensor was humidity- or composition-corrected are not stated - charcoal and tobacco smoke is exactly where optical PM monitors read high; five venues, no personal sampling, no worker shift-length exposure. Relevance: high for occupational indoor monitoring, where a factor-of-two optical bias still leaves concentrations an order of magnitude above guideline.

doi: [10.1155/ina/8494753](https://doi.org/10.1155/ina/8494753)**Campiotti et al. 2026** Build Environ*Beyond leaf traits: context dependent particulate matter accumulation across outdoor, indoor and active green walls*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 29 Sep with no abstract; PubMed efetch, Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates a comparison of leaf PM accumulation across outdoor, indoor and actively ventilated green walls. Relevant to vegetation-based mitigation claims, which need concentration measurements, not leaf loading, to show an exposure benefit. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: [10.1016/j.buildenv.2026.115347](https://doi.org/10.1016/j.buildenv.2026.115347)**Charres et al. 2026** Atmos Pollut Res*Insights into the organic and toxicological characteristics of PM_{10} from indoor and outdoor settings in a school environment*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 2 Oct with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes organic and toxicological characterisation of indoor and outdoor PM_{10} at a school; the author list (Cipoli, Vicente, Evtugina) is the Aveiro aerosol group, but this watch does not infer location or method from authorship. Relevance: potentially high - indoor/outdoor school PM composition is what classroom low-cost sensors cannot report. On the retry list.

doi: [10.1016/j.apr.2026.103223](https://doi.org/10.1016/j.apr.2026.103223)**Dennis & Ball 2026** AORN J*Strategies to Improve Compliance With Smoke Safety.*

Nursing practice review on surgical-smoke evacuation compliance: smoke from energy-based devices carries toxic gases, particulate matter and potentially viable cells; perioperative nurses report twice the general-population incidence of respiratory problems; barriers are education, equipment and culture; recommended strategies are gap analysis, training, leadership support and legislation. No exposure measurement or new data. Relevance: low; OR smoke is a high-concentration ultrafine source for which real-time particle counters would make evacuation compliance auditable.

doi: [10.1002/aorn.70159](https://doi.org/10.1002/aorn.70159)**Eghomwanre & Akharame 2026** Environ Sci Pollut Res Int*Spatio-temporal assessment of classroom air quality and thermal comfort as determinants of respiratory health risks in urban pre-tertiary institutions.*

Ten weeks of classroom monitoring ($PM_{2.5}$, PM_{10} , CO, CO_2 , thermal indices) in six primary and secondary schools in Benin City, with modified BMRC respiratory questionnaires. $PM_{2.5}$ ranged $11.2-81.4$ and PM_{10} $20.1-138.6 \mu g m^{-3}$; CO_2 peaked at 729.3 ppm; THI 28.3-30.1. HQs were all <1, yet cough (58.2%), phlegm (32.5%) and chest pain (26.7%) were common; higher PM was associated with cough ($p < 0.001$) and chest pain (crude OR 2.74), while poor sanitation predicted phlegm (aOR 5.54, 95% CI 1.35-22.80). Instrument type, calibration and student n are not in the abstract, and the PM ORs are crude. Relevance: West African classroom data are scarce; this is the setting where a school sensor network would be the only monitoring.

doi: [10.1007/s11356-026-38254-1](https://doi.org/10.1007/s11356-026-38254-1)**Gulzar et al. 2026** Atmos Environ*PAHs Exposure, Health Risk Evaluation, and Renal Associations by Household Fuel Type in Rural Residents of Northwestern China*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 28 Sep with no abstract; PubMed efetch, Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates PAH exposure, risk assessment and renal associations by household fuel type among rural residents of northwestern China. Pairs with the 25 Sep household solid-fuel records; retry for the exposure matrix (air, urine) and renal endpoint. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: [10.1016/j.atmosenv.2026.122388](https://doi.org/10.1016/j.atmosenv.2026.122388)**Konisky et al. 2026** Build Environ*Evaluating Indoor Air Pollution in Urban, Low-Income, Multi-Family Buildings: A Case Study from Indianapolis*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 1 Oct with no abstract; PubMed efetch, Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes an indoor air-pollution evaluation in low-income multi-family buildings in an Indianapolis case study. Directly relevant to low-cost indoor sensor deployments in under-monitored housing. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: [10.1016/j.buildenv.2026.115356](https://doi.org/10.1016/j.buildenv.2026.115356)

Liu et al. 2026 Environ Res

Spatiotemporal Contamination Characteristics, Bacterial Community Structure, and Airflow-Mediated Transport of Bacterial Aerosols in a Pediatric Emergency Consulting Room.

One paediatric emergency consulting room: culturable airborne and surface bacteria, high-throughput sequencing and CFD of aerosol transport and deposition. Airborne culturable counts differed little between locations (well-mixed small room), surface deposition was heterogeneous, and 73.47% of bacterial aerosol was below 3.3 µm; Staphylococcus was the core genus. CFD placed the consultation desk and floor at 79.71% of deposited particles, with high-risk zones around occupants. One room, culturable fraction only (typically <1% of total bacteria), and sampling duration and counts are not in the abstract. Relevance: marginal; bioaerosol is invisible to optical PM sensors, so ventilation decisions here need culture or fluorescence methods.

doi: 10.1016/j.envres.2026.125789

Liu et al. 2026 Atmosphere

Research on Comprehensive Evaluation of Air Quality in Underground Stations of Urban Rail Transit Systems

One island-platform Beijing Metro station: CFD of airflow for train absent, arriving and departing was used to place measurement points, then handheld instruments measured CO₂, PM_{2.5}, PM₁₀ and TVOC at morning and evening peaks, combined in an entropy-weighted index. Retention zones formed at the platform centre and escalator entrances under low-frequency ventilation; piston flow improved mixing but imported tunnel air; PM tracked outdoor inputs with some internal contribution. No concentrations, instrument models or sampling durations are in the abstract. Relevance: CFD-guided siting is directly transferable to placing fixed low-cost PM nodes in stations, where iron-rich wear particles bias optical readings.

doi: 10.3390/atmos17100938

Parthiban et al. 2026 Cureus

Ocular Surface Disorders in Traffic Police Officers in the Union Territory of Puducherry, India: A Cross-Sectional Study

106 traffic police examined with OSDI, Schirmer, TBUT, tear meniscus height and conjunctival impression cytology: Nelson grade 1-3 metaplasia in 83%, dry-eye symptoms in 78%; grade associated with years of field duty, OSDI and TBUT (chi-square p<0.05). Limitation: no unexposed comparison group, no exposure measurement, UV and pollution are inseparable, and years of service is collinear with age; the authors' 'clear exposure-outcome relationship' is not supported by this design. Relevance: low - a case for wearable PM monitoring in traffic-police cohorts, which would make this question answerable.

doi: 10.7759/cureus.115533

Rodriguez et al. 2026 Indoor Air

Integrated Assessment of Indoor Air Pollution and Associated Health Risks in a Hemodialysis Unit

One hemodialysis treatment room monitored for a year: ~1,400 time-resolved readings of CO₂, O₃ and PM_{0.3/2.5/10} with simultaneous outdoor measurements, plus 12 monthly VOC campaigns. Indoor CO₂ often exceeded the Spanish threshold (indoor-outdoor difference >350 ppm, peaks >1,100 ppm), indicating poor ventilation. PM_{2.5}, PM₁₀ and O₃ had I/O ratios below 1 (predominantly outdoor origin) and were generally under WHO guidelines with occasional peaks; most VOCs had I/O >1, and benzene and ethylbenzene cancer risks were below 1e-6. Limitation: one room, instrument type and PM calibration are not given in the abstract, and 'occasional peaks' are not quantified. Relevance: moderate - dialysis patients spend ~12 h/week indoors in one room; I/O ratios tell a facility whether filtration or ventilation is the lever.

doi: 10.1155/ina/4283416

Wang et al. 2026 Toxics

Mixed Exposure to Underground Air Pollutants and Metabolic Syndrome in Coal Miners: A Cross-Sectional Study Integrating Multi-Pollutant Models and Urinary Metabolomics.

Cross-sectional study of underground coal miners with logistic regression, BKMR and Qgcomp for coal dust, CO, CO₂ and NO₂, plus untargeted urinary metabolomics. Each pollutant was associated with metabolic syndrome; the mixture was positively associated in BKMR and Qgcomp, with coal dust and NO₂ contributing most - dust tracking central obesity and hypertension, NO₂ glucose and triglycerides. >113 metabolites differed; acylcarnitines tracked dust and leukotrienes NO₂. The abstract gives no n, no effect sizes and no exposure-measurement method; healthy-worker and shift-work confounding are unaddressed. Relevance: adds a metabolic endpoint to respirable coal dust, which is measured underground by personal dust monitors that low-cost sensors cannot yet replace.

doi: 10.3390/toxics14090746

Yu et al. 2026 Atmosphere

Development of a Qualitative Questionnaire for Indoor Air Quality Monitoring: A Scoping Review of Subjective Assessments in Studies Using Low-Cost Sensors

Scopus-only scoping review (281 records, 35 included) of questionnaires used alongside low-cost IAQ sensors in homes and workplaces, grouping items into five domains - demographics, symptoms, perception, building/occupancy, monitored pollutants - and proposing a structured questionnaire. A single database and no appraisal of questionnaire validity limit it to an item inventory; no psychometric testing of the proposed tool is reported. Relevance: moderate for deployment design - pairing sensor time series with standardised perception items is how sensor data become usable for SBS-type outcomes.

doi: 10.3390/atmos17100943

Yuksel & Yildirim 2026 Front Public Health

An analysis of occupational exposure to particulate matter and erythrocyte morphological abnormalities among charcoal-grill restaurant workers

25 charcoal-grill restaurant workers vs 25 office staff. Kitchen PM median 117.8 vs 0.23 µg m⁻³ (the office value is implausibly low for any indoor space and suggests an instrument floor), with more wheeze (12/25 vs 0/25). After adjustment, four poikilocyte types were elevated - dacryocytes IRR 2.46 (95% CI 1.36-4.44), elliptocytes 2.33, sickle forms 4.47, schistocytes 1.56 - but total anomaly burden was not (IRR 1.57, 0.65-3.78). n=25 per arm, smears only for 19 controls, CO co-exposure unseparated and smoking unreported. Relevance: low-moderate - one more occupational combustion setting with nine-fold-plus PM gradients that cheap sensors could document over full shifts.

doi: 10.3389/fpubh.2026.1953388

Burden, policy & mitigation**14 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (14 OF 14 RECORDS IN THIS SUBTOPIC)

Bong et al. 2026 medRxiv (preprint)*Delayed Coal Plant Retirements in the Era of Data Centers Carry a Measurable Mortality Cost*

Of 50 US coal plants scheduled (Jan 2024) to retire before 2035, 20 had later or removed dates by Jan 2026. Projecting plant-attributable PM_{2.5} and mortality among Medicare beneficiaries for 2026-2035, the revised schedule adds 1,256 deaths (1,136-1,376), 3.1 times the burden of the original schedule. Eighteen of the 20 plants sit in balancing authorities with hyperscale data-centre demand, and several federal orders and utility filings cite data-centre growth. Limitation: the interval reflects concentration-response uncertainty only, not dispersion-model or future-emission uncertainty; operating hours of delayed plants are assumed; attribution to data centres is circumstantial, not causal; preprint. Relevance: moderate - coal-sulfate PM_{2.5} is regional and secondary, so it is a background term networks see as a baseline, not as a local plume.

doi: 10.64898/2026.09.29.26364336

Busskamp et al. 2026 Atmos Environ*A virtual limit value for ultrafine particles: Frequency of threshold exceedances across Germany based on data from the German Ultrafine Aerosol Network*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 29 Sep with no abstract; PubMed efetch, Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates a test of a hypothetical ultrafine-particle limit value against German Ultrafine Aerosol Network data. High relevance: the revised EU Ambient Air Quality Directive requires UFP monitoring without a limit value, and any future threshold will define what number-counting sensors must resolve. Retry first. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122390

Cheong et al. 2026 J Expo Sci Environ Epidemiol*Community-level air quality and health implications of sustainable aviation fuels adoption: a scoping review*

METADATA ONLY. Metadata-only record: Springer Nature deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes a scoping review of near-airport air quality and health under sustainable aviation fuel adoption. Relevant to ultrafine-particle monitoring around airports. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1038/s41370-026-00971-7

Jin Z. et al. 2026 Environ Sci Technol*Carbonaceous Components Underlying Wildfire Smoke PM_{2.5}-Associated Mortality in Older U.S. Adults*

77.5 million Medicare beneficiaries (2002-2019), annual high-resolution smoke and non-smoke PM_{2.5} and carbonaceous matter, stratified Cox models. Per 1 $\mu\text{g m}^{-3}$: HR 1.032 (95% CI 1.027-1.037) for smoke PM_{2.5} and 1.087 (1.076-1.098) for smoke carbonaceous matter, both above the non-smoke counterparts per unit mass; per IQR the ranking reverses for mass because smoke exposure varies little. Estimated 22,420 deaths/yr attributable to smoke PM_{2.5}, mostly its carbon. Stronger in Black and dual-eligible beneficiaries. Annual means smooth episodic smoke into a chronic metric, and smoke attribution is itself modelled. Relevance: high - composition-resolved smoke exposure is the case for adding absorption/BC channels to LCS networks in fire regions.

doi: 10.1021/acs.est.6c07219

Lee et al. 2026c ACS Omega*Exhaust Emission Characteristics of Gasoline Engines with Secondary Air Systems and Electrically Heated Catalysts in Cold-Start WLTC Mode*

A 1.6 L GDI engine with warm-up and underfloor catalysts, an electrically heated catalyst (EHC) and secondary air injection (SAI) was run over the cold WLTC phase, including a -7 C coolant case. EHC + SAI over the first 80 s cut THC 69.1%, CO 65.7% and NO_x 80.6% for 26.7 Wh; best strategy was EHC pre-heat 50 s before start, held 30 s after. SAI without EHC raised CO. PN rose sharply in the cold case but optimised EHC-SAI coordination cut it by >30%. Limitation: one engine, one calibration, dynamometer cycle rather than real driving, and PN is reported only as a relative reduction - no absolute PN, size distribution or sub-23 nm fraction, which is where cold-start GDI particles sit. Relevance: moderate - cold-start PN is the largest remaining tailpipe contribution in urban low-cost-network data, and sub-23 nm particles are invisible to the optical nodes watching for it.

doi: 10.1021/acsomega.6c05980

Loreses Noguera & Cardenas-Duran 2026 QJM*Disentangling Population Ageing from Epidemiological Change in the Rising Burden of PM_{2.5}-Attributable Non-Communicable Disease*

METADATA ONLY. Metadata-only record: Oxford University Press deposited title, authors and DOI on 28 Sep with no abstract; PubMed efetch, Crossref, Europe PMC and the publisher landing page returned nothing. The title indicates a decomposition of the rising PM_{2.5}-attributable NCD burden into population ageing versus epidemiological change; an author response to correspondence was deposited the same day. The ageing-versus-exposure split is the same point the 25 Sep Wan et al. ES&T projection made for China. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1093/qjmed/hcag257

Ma et al. 2026 Environ Sci Technol*Afforestation and Emission Cuts: Dual Drivers for China's Clean Air and Carbon Neutrality*

Emission inventories plus a chemical transport model test maximum-potential afforestation against deep anthropogenic cuts. Afforestation raises summer BVOC emissions by 3.20-3.82 Tg (20-24%), adding 0.39 $\mu\text{g m}^{-3}$ PM_{2.5} (range 0.02-3.47) and 0.74 ppbv MDA8 O₃ in eastern China under current emissions; under carbon-neutral emissions SOA-related PM_{2.5} from afforestation falls 67% and the O₃ response turns slightly negative. Species choice alone can cut local O₃ impact to under a third. BVOC emission factors and SOA yields are the dominant, unquantified uncertainty. Relevance: low-moderate - the PM_{2.5} penalty is small relative to sensor error.

doi: 10.1021/acs.est.6c04722

Manchanda et al. 2026 Environ Sci Technol*Inverse Modeling Identifies Efficient Emission Control Strategy for Mitigating PM_{2.5} Pollution*

Recasts attainment planning as a constrained Bayesian inverse problem: find the smallest spatially explicit, sector- and precursor-resolved emission changes that bring modelled PM_{2.5} under the 9 $\mu\text{g m}^{-3}$ annual NAAQS across the contiguous US. Strategies with similar total reductions ranged from full compliance to minimal progress depending on where and what was cut; aggregate tonnage does not predict attainment. No numeric results in the abstract; the solution inherits the transport model's linearised source-receptor relationships and inventory error. Relevance: moderate - attainment is judged at monitors, so where the monitors (and dense sensor networks) are determines which 'efficient' strategy appears to work.

doi: 10.1021/acs.est.6c06581

Pannu et al. 2026 Research Square (preprint)

Assessment of the significance of urban vegetation in air phytoremediation: Quantitative analysis of particulate matter removal

Leaves of five morphologies were sampled every fourth day for a year, leaf-surface PM quantified in three size classes and scaled to 20 dominant species and canopy cover. Delhi's urban forest (23% of area) removed 527 +/- 301 t PM_{2.5} and 7,907 +/- 1,079 t total PM - about 1.08% of annual PM_{2.5} emissions in the 2018 SAFAR inventory. Directly measured removal is ~24% of the deposition-velocity estimate for PM_{2.5} and ~4.6% for PM₁₀; leaf loading saturates in Nov-Jan. Fine deposits are enriched in Cr, Ni, Cu, Zn and Pb. Limitation: five morphologies extrapolated to 20 species, wash-off and resuspension partly estimated, an emissions denominator from 2018; preprint. Relevance: moderate - urban greening claims based on deposition-velocity models overstate removal four-fold.

doi: 10.21203/rs.3.rs-10839085/v1

Quadros et al. 2026 Commun Earth Environ

Global air quality and human health impacts of growing aircraft emissions

Global aircraft emissions projected 2019-2040 and run through a chemistry transport model. 2019: 33,900 (95% CI 23,500-45,600) PM_{2.5} deaths and 24,600 (15,500-34,200) O₃ deaths attributable to aviation. By 2040 these rise 58%/82% (low), 119%/148% (baseline) and 169%/205% (high scenario); the non-aviation background shifts aircraft impacts by up to 44%. Limitation: the CIs propagate concentration-response uncertainty only, not CTM, inventory (especially cruise NOx-nitrate) or grid resolution, which dominate a marginal-sector estimate; most PM_{2.5} impact is secondary nitrate and sulfate far from airports. Relevance: moderate - the near-airport UFP signal low-cost networks can see is not the mass-based burden counted here.

doi: 10.1038/s43247-026-03732-4

Wang et al. 2026b J Environ Sci (China)

What drives long-term variations in biomass burning emissions in China?

A 2008-2023 inventory of domestic and open biomass burning with Kaya/LMDI decomposition of PM_{2.5} emission change. 2008 emissions were 3.22 Mt PM_{2.5}, 0.50 Mt EC and 1.41 Mt OC; all species fell 50-65% by 2023. Domestic burning declined mainly through the economic factor on burned biomass (183.4% of the change) and urbanisation (54.7%); open straw burning fell through activity intensity (-115%), whose effect weakened after 2017. Limitation: LMDI partitions an identity, it does not test causation, and activity data for domestic and open burning are the weakest inputs of any Chinese inventory; no uncertainty bounds are given. Relevance: low-moderate - the declining rural smoke signal is a baseline shift any rural sensor trend analysis has to account for.

doi: 10.1016/j.jes.2026.05.013

Yalali et al. 2026 Front Public Health

Geospatial modeling of environmental and socioeconomic drivers of health outcomes.

Census-tract prevalence of seven chronic conditions (hypertension, asthma, COPD, stroke, cancer, diabetes, mental distress) in Harris County, Texas, modelled with OLS, GWR and multiscale GWR on 2022 data and re-estimated for 2020-2024. MGWR raised adjusted R² (e.g. 0.57 to 0.75 for hypertension) and poverty and PM_{2.5} were the most consistent predictors, strongest near the eastern industrial corridors; patterns were stable across years (|dR²| <= 0.09). Outcomes are modelled small-area estimates (likely CDC PLACES), so the regression partly recovers the covariates used to make them; no effect size for PM_{2.5} is given. Ecological. Relevance: low; locates tracts where added sensors would coincide with high modelled burden.

doi: 10.3389/fpubh.2026.1790559

Yang et al. 2026b Environ Res

Process-resolved characterization and plant-level inventory of multi-pollutant emissions from China's coking industry

Stack measurements at coal charging, coke-oven flue gas, dry quenching and pushing in two plants (total PM 3.12-37.72 mg/m³; sulfate 28-59% of ions; As and Cd enriched in PM_{2.5}; the condensable fraction carries 44.5-57.4% of trace elements in flue gas) feed a 307-plant inventory for 2012-2024. 2022 national emissions: 159.8 kt PM, 66.6 kt SO₂, 82.3 kt NO_x, 838.3 kt VOCs, 4.24 kt PAHs, 707.6 t trace elements; BACT cuts PM to 15.5 kt by 2050. Limitation: two plants anchor factors for 307; fugitive emissions, flagged as a priority, are the least measured; condensable PM is operationally defined. Relevance: moderate - condensables are invisible to stack filters but appear as ambient PM_{2.5} downwind, where fence-line sensor networks see them.

doi: 10.1016/j.envres.2026.125824

Zhang et al. 2026d Atmos Pollut Res

Impact of strong biomass burning in Northeast China: air pollution, enhancement ratio among pollutants, and health risk

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 3 Oct with no abstract. By title, an episode study of strong agricultural biomass burning in Northeast China reporting enhancement ratios among co-emitted pollutants and a health-risk estimate. Enhancement ratios (dPM / dCO, for instance) are the quantity that lets a sparse network attribute an episode to burning rather than to secondary formation, so the numbers matter more than the burden claim. Relevance: moderate; on the retry list.

doi: 10.1016/j.apr.2026.103224

Other clinical endpoints

5 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (5 OF 5 RECORDS IN THIS SUBTOPIC)

Gyeltshen et al. 2026 Int J Cancer

Short-Term PM_{2.5} Exposure and Daily Cancer Mortality Across 47 Japanese Prefectures: A Nationwide Time-Stratified Case-Crossover Study, 2013-2022

Nationwide case-crossover, all 47 prefectures 2013-2022, prefecture-specific conditional quasi-Poisson pooled by random effects. Per 10 µg m⁻³ PM_{2.5} at lag 0-2, all-cancer mortality rose 0.88% (95% CI 0.57-1.18), with breast 2.06%, colorectal 1.06%, stomach 0.81% and lung 0.77%, and no effect modification by sex, age or site. The association faded beyond lag 0-5 and lost significance after adjustment for NO₂ or SO₂ - the decisive weakness: an acute cancer-death effect is most plausibly harvesting of frail terminal patients via cardiorespiratory pathways, and co-pollutant sensitivity leaves PM specificity open. Prefecture-mean exposure. Relevance: moderate - prefecture-scale exposure is coarse; finer networks would test the co-pollutant question.

doi: 10.1002/ijc.70761

Lima et al. 2026 medRxiv (preprint)*Epigenetic Exposure Signatures Distinguish Early-Onset from Later-Onset Colorectal Cancer in a Racially Diverse US Cohort*

Applies published methylation risk scores (lifestyle, air pollution, pesticides) to tumour tissue from 146 DANCE colorectal-cancer cases (29 early-onset, 69 intermediate, 48 late) and relates them to age at onset, adjusting for race, geography, Area Deprivation Index and a multi-exposure index. Higher odds of early onset were associated with MRS for 2,4-D, heptachlor, coarse PM, Mediterranean diet and smoking. Limitation: decisive - MRS were trained on blood, applied here to tumour DNA, whose methylome is reshaped by the cancer itself; 29 early-onset cases; signatures are proxies validated against observed exposure only by Spearman correlation; no effect sizes in the abstract; preprint. Relevance: low - a coarse-PM methylation proxy is not an exposure measurement.

doi: 10.64898/2026.09.28.26363311

Sandal et al. 2026 Clin Kidney J*Disasters and hazard exposure in end-stage kidney disease: a scoping review of patient outcomes*

PRISMA scoping review: 10,040 screened, 30 studies, ~1.07 million dialysis patients and 64,531 transplant recipients. Poor air quality was associated with 5-139% higher mortality; extreme heat 9-45%; 23-59% of patients had dialysis disruption in disasters. Limitation: scoping design, no pooling or risk-of-bias grading, and 'poor air quality' spans wildfire smoke, PM_{2.5} increments and AQI categories, so the 5-139% range is not an effect size. Relevance: low-moderate - dialysis patients are an identifiable, routinely contacted population where sensor-triggered alerts could plausibly be tested.

doi: 10.1093/ckj/sfag318

Xia et al. 2026 J Hazard Mater*Long-term exposure to fine particulate matter constituents and lung cancer risk in Chinese never-smokers*

262,785 never-smokers in the China Kadoorie Biobank, 2,646 incident lung cancers; time-dependent Cox models per IQR of each constituent, quantile g-computation for the mixture, and a polygenic risk score for interaction. HR per IQR: nitrate 1.35 (1.23-1.48), ammonium 1.22 (1.13-1.31), chloride 1.20 (1.12-1.28); mixture HR 1.13 (1.07-1.19), dominated by NO₃- and NH₄+; additive and multiplicative NO₃- x PRS interaction. Limitation: constituents come from one satellite/ML reanalysis in which nitrate and ammonium are strongly collinear and partly inherited from total-mass predictors, so 'nitrate drives the risk' may be 'the best-resolved component wins'; second-hand smoke and cooking fuel only partly captured. Relevance: high - constituent-specific risk is the argument for composition-capable nodes (nitrate is semi-volatile and biased low on heated optical sensors).

doi: 10.1016/j.jhazmat.2026.143759

Yang and Liu 2026 medRxiv (preprint)*No evidence for large causal effects of six air pollutants on non-suppurative otitis media: a two-sample Mendelian randomization study*

Two-sample MR with UK Biobank pollutant GWAS and FinnGen R13 OME (12,397 cases, 464,237 controls); IVW primary, full pleiotropy battery, positive (asthma, rhinitis, CRS) and negative (suppurative OM) controls. No pollutant survived FDR; the nominal PM₁₀ OR 1.56 (0.95-2.55) reversed with the 88-SNP set (0.64, 0.24-1.69). Minimum detectable ORs 1.69-32.1, so moderate effects are not excluded - the authors say so. Limitation: 'genetic instruments for ambient PM' index residential and behavioural traits, not exposure; null MR here is uninformative by construction, and MVMR F < 10. Relevance: low - included as a properly bounded null, not as evidence of absence.

doi: 10.64898/2026.09.30.26364391

Pooled corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Bloomberg et al.	Daily Exceedance of World Health Organization Air Pollutant Guidelines and Cognitive Aging	medRxiv (preprint)	Prospective cohort	Annual count of days above WHO daily guideline for PM _{2.5} , PM ₁₀ , NO ₂	United Kingdom (England, ELSA)
Han et al.	Air pollution and criminal behavior: a nationwide case-crossover analysis in China	Int J Epidemiol	Time-stratified case-crossover	Daily PM _{2.5} , PM ₁₀ , NO ₂ , CO, SO ₂ , O ₃ (six pollutants, WQS mixture)	China (national)
Januel et al.	Air pollution and the risk of multiple sclerosis relapses in the Paris region, the CONFISEP study	J Neurol Neurosurg Psychiatry	Self-controlled design	Weekly street-level PM ₁₀ , PM _{2.5} , NO ₂ , O ₃ (Airparif model)	France (Ile-de-France)
Kwon	Fine particulate matter (PM _{2.5}) and suicide in South Korea: an instrumental variable analysis using thermal inversion.	Front Public Health	Instrumental-variable panel analysis	Regional PM _{2.5} instrumented by thermal inversions, 2015-2019	South Korea
Shupler et al.	Preconception, Prenatal and Postnatal PM _{2.5} Exposure and Neurodevelopmental Disorder Risk among Medicaid Recipients	Environ Sci Technol	Retrospective cohort	ZIP-code PM _{2.5} , ZIP-code PM _{2.5} , PM ₁₀ , NO ₂ , CO, SO ₂ , O ₃ (six pollutants, WQS mixture)	USA (Medicaid, national)
Wang et al.	Environmental exposures and mental health: longitudinal results from the German National Cohort	Environ Int	Metadata only (no abstract)	Metadata only - environmental exposures and mental health in NAKO (by title)	Not recoverable from metadata

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
Cosenza et al.	Associations of Outdoor Air Pollution With Cause-specific Mortality in Metastatic Prostate Cancer: A cohort study	<i>medRxiv (preprint)</i>	Retrospective cohort	Residential PM _{2.5} and NO ₂ at diagnosis, 1-km exposure models	USA (11 state registries)
Jin et al.	Ambient air pollution and stroke in two population-based cohorts: roles of systemic inflammation and polygenic risk.	<i>Neuroepidemiology</i>	Prospective cohort	Residential PM _{2.5} , PM ₁₀ (CHARLS; UK Biobank LUR)	China and UK (CHARLS, UK Biobank)
Kriti et al.	Ecological associations between ambient air pollution, gut microbiome composition, and metabolic markers in urban populations of central India.	<i>Front Public Health</i>	Ecological	Station PM _{2.5} , PM ₁₀ , NO _x , SO ₂ by residential locality	India (Bhopal)
Lin et al.	Short-term lagged exposure to ambient air pollutants and hospitalization costs among AMI patients: health-economic evidence from a subtropical city	<i>Front Public Health</i>	Retrospective cohort	Lag 0-6 day PM _{2.5} , PM ₁₀ , SO ₂ , NO ₂ , CO, O ₃ (city monitors)	China (Jiangmen, Guangdong)
Rivera-Mariani et al.	Inflammatory Burden, Structural Exposure, and Racial Disparities in Incident Heart Failure: Evidence from the Multi-Ethnic Study of Atherosclerosis	<i>medRxiv (preprint)</i>	Prospective cohort	PM _{2.5} and NO ₂ as co-adjustment in a geocoded subcohort (not the exposure of interest)	USA (MESA)
Sun et al.	Solid Fuel Use and Risks of All-Cause and Cardiovascular Disease Mortality in Urban Northern China: A 12-Year Cohort Study	<i>Indoor Air</i>	Retrospective cohort	Household solid fuel use for heating/cooking (years); no PM measured	China (Tianjin, Shenyang, Taiyuan, Rizhao)
Wang et al.	Long-term PM _{2.5} exposure and incident heart failure in the UK Biobank: relative and absolute risk estimates.	<i>Front Public Health</i>	Prospective cohort	LUR annual PM _{2.5} , PM ₁₀ , NO ₂ (2010 ESCAPE surface)	UK
Yu et al.	Effect modification of greenness on the associations between air pollutants and out-of-hospital cardiac arrest: Evidence from a multi-city study	<i>Public Health</i>	Time-stratified case-crossover	Short-term PM _{2.5} , PM ₁₀ , O ₃ , CO at onset location; NDVI/EVI at 250-1000 m	China (multi-city)
Reproductive & developmental					
Bae et al.	Exposure to ambient air pollutants during early gestation and phenotypic variation in non-syndromic microtia.	<i>J Plast Reconstr Aesthet Surg</i>	Observational - cross-sectional	Modelled PM _{2.5} , PM ₁₀ , O ₃ , NO ₂ , SO ₂ , CO at gestational weeks 5-9 (address at registration)	South Korea (single centre)
Gielen et al.	The association of neighborhood socioeconomic status with birth weight is not confounded by ambient air pollution in twin pregnancies	<i>Neonatology</i>	Prospective birth cohort	Kriged PM ₁₀ and NO ₂ at residence (monitor + land cover)	Belgium (East Flanders)
Ha et al.	Developmental PM _{2.5} exposure and stage-dependent kidney senescence	<i>Kidney Res Clin Pract</i>	Metadata only (no abstract)	Metadata only - developmental PM _{2.5} exposure (by title)	Not recoverable from metadata
Kukshal et al.	Maternal Exposure to Air Pollution During Early Pregnancy and Congenital Heart Disease: An Exploratory Case-Control Analysis from Delhi NCR	<i>Indian J Pediatr</i>	Case-control	Periconceptional PM _{2.5} , PM ₁₀ , NO ₂ , CO, SO ₂ , O ₃ , AQI at geocoded residence	India (Delhi NCR)
Ma et al.	Maternal PM _{2.5} exposure induces preeclampsia-like phenotypes in mice through placental angiogenic dysfunction and lncRNA-RUNX1 dysregulation	<i>J Environ Manage</i>	Animal model	Heating-season northern-China PM _{2.5} , intranasal 3 mg/kg every other day, GD0.5-17.5	China (laboratory; Taiyuan-type heating-season PM)
Song et al.	Role of Neighborhood Vulnerability and Opportunity in the Association Between Prenatal Air Pollution and Autism-Related Traits	<i>Environ Res</i>	Prospective birth cohort	Prenatal PM _{2.5} , NO ₂ , O ₃ at residence (pregnancy and trimester means)	USA (ECHO, 35 sites)
Wang et al.	Long-term Exposure to PM _{2.5} and Risk of Adverse Pregnancy Events in Women with Pulmonary Hypertension: A Nationwide Multicenter Study	<i>J Heart Lung Transplant</i>	Retrospective cohort	Monthly PM _{2.5} , 3-year mean before end of pregnancy; trimester windows	China (24 tertiary hospitals)

First author	Focus	Journal	Design	Metric	Region
Zhang et al.	Second-trimester PM _{2.5} and PM ₁₀ exposure is associated with neonatal metabolome alterations and elevated risk of pregnancy loss	<i>Environ Pollut</i>	Prospective cohort	PM _{2.5} and PM ₁₀ in seven windows, pre-pregnancy to peripartum; neonatal metabolites	China
Respiratory & allergic					
Carroll et al.	Global Tuberculosis Burden Attributable to Household and Outdoor Air Pollution	<i>Ann Am Thorac Soc</i>	Systematic review / meta-analysis	Household solid fuel; outdoor PM _{2.5} (short- and long-term)	Global / multi-region (review)
Kulkarni et al.	Short-Term Exposure to PM _{2.5} and Emergency Department Visits for Asthma and Chronic Obstructive Pulmonary Disease.	<i>JAMA Netw Open</i>	Metadata only (no abstract)	Short-term ambient PM _{2.5} (details not deposited)	Singapore
Maunakea et al.	Climate-amplified wildland-urban interface disasters create persistent internal metal mixtures associated with pulmonary impairment.	<i>Proc Natl Acad Sci U S A</i>	Observational - cross-sectional	24 urinary metals 6-18 months after the Aug 2023 Maui WUI fires (internal dose, not air)	USA (Maui, Hawaii)
Shi et al.	Hidden Hazards at Home: Indoor Air Pollution and Adolescent Asthma - A Systematic Review and Meta-Analysis	<i>Indoor Air</i>	Systematic review + random-effects meta-analysis	Indoor exposures incl. cooking/heating combustion and open fires	Global / multi-region (review)
Zouma et al.	Pulmonary function assessment of a community exposed to air pollution in an industrial area in Ouagadougou, Burkina Faso	<i>Toxicol Rep</i>	Observational - cross-sectional	Residence in an industrial area vs control district (no PM measured in abstract)	Burkina Faso (Ouagadougou)
Mechanistic toxicology					
Cai et al.	Prenatal PM _{2.5} exposure disrupts cortical excitatory/inhibitory balance in male offspring by suppressing Lhx8-regulated cholinergic homeostasis and cell adhesion	<i>Ecotoxical Environ Saf</i>	Animal experiment	Prenatal PM _{2.5} (mouse model; dose not in abstract)	Not stated (laboratory)
Driencourt et al.	The impact of post-exposure Streptococcus pneumoniae infections on pulmonary immunity: The role of air pollution and IL-20 cytokines in respiratory infections.	<i>Rev Mal Respir</i>	Narrative review	Traffic-derived PM _{2.5} (conceptual)	France
Ferreira Vicente et al.	Pulmonary black carbon accumulation: mechanisms, health impacts, and environmental sources	<i>Expert Rev Respir Med</i>	Narrative review	Black carbon: sources, tissue retention, label-free optical detection	Not stated (review)
Jiang et al.	Identification of shared and divergent molecular pathways linking ambient air pollutants to mortality and major chronic diseases through proteomic profiling	<i>Redox Biol</i>	Cohort proteomics + mediation + Mendelian randomization	Residential PM _{2.5} , PM ₁₀ , NO ₂ , NO _x (LUR)	United Kingdom (UK Biobank)
Jin et al.	ANKRD22 contributes to NF-κB-associated inflammatory responses in alveolar type II epithelial cells during PM _{2.5} -triggered airway inflammation	<i>Front Immunol</i>	Animal + in vitro	PM _{2.5} instillation (mouse) and A549 exposure; dose not in abstract	China (laboratory)
Kamyabi et al.	A novel controlled wood smoke exposure model to evaluate physiological responses to smoke-derived air pollution in dairy calves.	<i>J Dairy Sci</i>	Animal experiment	6-h wood smoke in enclosures: PM _{2.5} 238.44 +/- 62.97 vs 9.94 +/- 4.50 µg/m ³	USA
Kim et al.	Acorus gramineus Solander Preserves Corneal Epithelial Barrier Integrity against PM _{2.5} -Induced EMT-Like Remodeling	<i>J Pharmacopuncture</i>	In vitro	PM _{2.5} on human corneal epithelial spheroids; dose not in abstract	South Korea (laboratory)
Lee et al.	Curcumin-Loaded Nanospheres Attenuate Fine Dust (PM ₁₀)-Induced Cellular Toxicity by Suppressing Reactive Oxygen Species-Dependent Signaling in Human Keratinocytes	<i>Cell Biol Int</i>	In vitro exposure	PM ₁₀ applied to HaCaT keratinocytes; curcumin nanospheres as the test article	South Korea (laboratory)
Liu et al.	Bulk PM _{2.5} Masks Oxidative Potential and Reactive Oxygen Species Responses by Metal-Organic Complexation in PM1 and PM1-2.5 Mixtures	<i>Environ Sci Technol</i>	Chamber / laboratory	Roadside PM1 and PM1-2.5 extracts; OP (AA, DTT), cellular ROS, mitochondrial superoxide	China (road-side site)

First author	Focus	Journal	Design	Metric	Region
Liu et al.	Metabolite-Associated Inflammation during Combined Exposure to Cooking Oil Fumes and Atopic Dermatitis: A Multi-Organ Spatial Metabolomics Study	<i>Environ Sci Technol</i>	Animal model	Cooking oil fume inhalation (controlled high-concentration regimen)	China (laboratory)
Onoda et al.	Gene-environment interaction in neurodevelopment: Setd5 haploinsufficiency and prenatal carbon nanoparticle exposure synergistically induce ASD-like phenotypes	<i>Part Fibre Toxicol</i>	Animal model	Carbon black nanoparticles, intranasal 95 ug/kg on gestational days 5 and 9	Japan (laboratory)
Saveleva et al.	Urban PM0.2 Affects Astrocytes by Inducing Xenobiotic and Oxidative Stress Responses	<i>Mol Neurobiol</i>	In vitro exposure	Urban ambient PM0.2 (aerodynamic diameter <0.2 um), 50 ug/ml, 24 h	Not stated (laboratory)
Zhang et al.	PM _{2.5} exposure impairs BMSC function and dysregulates mitochondrial oxidative stress: multi-omics insights into potential osteoporotic mechanisms	<i>Free Radic Biol Med</i>	In vitro exposure	PM _{2.5} suspension applied to rat bone-marrow mesenchymal stem cells (dose not in abstract)	China (laboratory)
Occupational & indoor					
Aljabri et al.	Indoor Air Quality and Exposure Characterization in Operational Waterpipe Cafes: Field Measurements of Particulate and Volatile Pollutants in Al-Madinah Al-Munawwarah	<i>Indoor Air</i>	Field measurement	Indoor PM _{2.5} and PM ₁₀ with TVOC and formaldehyde, peak occupancy, 5 venues	Saudi Arabia (Al-Madinah)
Campiotti et al.	Beyond leaf traits: context dependent particulate matter accumulation across outdoor, indoor and active green walls	<i>Build Environ</i>	Metadata only (no abstract)	Metadata only - PM deposition on outdoor, indoor and active green walls (by title)	Not recoverable from metadata
Charres et al.	Insights into the organic and toxicological characteristics of PM ₁₀ from indoor and outdoor settings in a school environment	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - indoor/outdoor school PM ₁₀ organic speciation and toxicity (by title)	Not recoverable from metadata
Dennis & Ball	Strategies to Improve Compliance With Smoke Safety.	<i>AORN J</i>	Narrative review	Surgical smoke (particles, gases) from energy-based devices	USA
Eghomwanre & Akharamé	Spatio-temporal assessment of classroom air quality and thermal comfort as determinants of respiratory health risks in urban pre-tertiary institutions.	<i>Environ Sci Pollut Res Int</i>	Comparative field monitoring (built environment)	Classroom PM _{2.5} , PM ₁₀ , CO, CO ₂ and thermal indices, six schools, 10 weeks	Nigeria (Benin City)
Gulzar et al.	PAHs Exposure, Health Risk Evaluation, and Renal Associations by Household Fuel Type in Rural Residents of Northwestern China	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - household-fuel PAH exposure and renal indicators (by title)	China (north-western, rural)
Konisky et al.	Evaluating Indoor Air Pollution in Urban, Low-Income, Multi-Family Buildings: A Case Study from Indianapolis	<i>Build Environ</i>	Metadata only (no abstract)	Metadata only - indoor air pollution in low-income housing (by title)	Not recoverable from metadata
Liu et al.	Spatiotemporal Contamination Characteristics, Bacterial Community Structure, and Airflow-Mediated Transport of Bacterial Aerosols in a Pediatric Emergency Consulting Room.	<i>Environ Res</i>	Repeated field measurement with CFD simulation	Size-resolved culturable bacterial aerosol (Andersen-type stages), CFD deposition	China (paediatric hospital)
Liu et al.	Research on Comprehensive Evaluation of Air Quality in Underground Stations of Urban Rail Transit Systems	<i>Atmosphere</i>	Repeated field measurement with CFD simulation	Handheld PM _{2.5} , PM ₁₀ , CO ₂ , TVOC at CFD-selected points; one island-platform station	China (Beijing Metro)
Parthiban et al.	Ocular Surface Disorders in Traffic Police Officers in the Union Territory of Puducherry, India: A Cross-Sectional Study	<i>Cureus</i>	Observational - cross-sectional	Occupational traffic exposure (>1 year field duty); no PM measured	India (Puducherry)
Rodriguez et al.	Integrated Assessment of Indoor Air Pollution and Associated Health Risks in a Hemodialysis Unit	<i>Indoor Air</i>	Field measurement	Indoor/outdoor PM0.3, PM _{2.5} , PM ₁₀ , O ₃ , CO ₂ ; monthly VOC campaigns	Spain (hemodialysis unit)

First author	Focus	Journal	Design	Metric	Region
Wang et al.	Mixed Exposure to Underground Air Pollutants and Metabolic Syndrome in Coal Miners: A Cross-Sectional Study Integrating Multi-Pollutant Models and Urinary Metabolomics.	<i>Toxics</i>	Observational - cross-sectional	Underground coal dust, CO, CO ₂ , NO ₂ (job-exposure assignment; method not in abstract)	China (coal mines)
Yu et al.	Development of a Qualitative Questionnaire for Indoor Air Quality Monitoring: A Scoping Review of Subjective Assessments in Studies Using Low-Cost Sensors	<i>Atmosphere</i>	Scoping review	Studies pairing low-cost IAQ sensors with occupant questionnaires	Global / multi-region (review)
Yuksel & Yildirim	An analysis of occupational exposure to particulate matter and erythrocyte morphological abnormalities among charcoal-grill restaurant workers	<i>Front Public Health</i>	Cross-sectional survey	Kitchen PM mass (spot), CO, CO ₂	Turkiye (Kocaeli)
Sensing, forecasting & instrumentation					
Aakko-Saksa et al.	Aggregated non-methane organic and inorganic gases as indicators of aged particulate matter induced by traffic	<i>Environ Sci: Atmos</i>	Metadata only (no abstract)	Metadata only - aggregate NMVOC as proxy for aged traffic PM (by deposit teaser)	Not recoverable from metadata
Chemyakin et al.	Single-scattering look-up table for remote sensing and in situ black carbon studies	<i>Opt Lett</i>	Algorithm development	Black carbon fractal aggregates, single-scattering properties 0.35-2.3 μm	Not stated (computational)
Chen et al.	Linking Aerosol Chemical Composition to Volatility: A Predictive Parameterization and Implications for Future Clean Air	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Metadata only - composition-to-volatility parameterisation (by title)	Not recoverable from metadata
Hall et al.	Indoor Air Quality Forecasting With Machine Learning: A Systematic Review	<i>Indoor Air</i>	Systematic review (PRISMA)	Indoor PM and other IAQ variables; forecasting models	Global / multi-region (review)
Herman et al.	An Advanced, Multifunctional, and Extreme-Condition Test Chamber for Performance Evaluation of Air Quality Sensors	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - sensor test chamber (by title)	Not recoverable from metadata
Hu et al.	Columnar optical properties of locally emitted light-absorbing carbonaceous aerosols	<i>J Environ Sci (China)</i>	Measurement campaign	BC and BrC absorption (AE33, 370/470 nm), AOD/AAOD/SSA at 550 nm	China (Beijing, surface + mountain station)
Jonsson et al.	Tracing biological, anthropogenic, and inorganic sources of coarse aerosols via single-particle fluorescence and optical morphology	<i>Atmos Chem Phys</i>	Laboratory instrument development	Coarse-mode single particles, Multiparameter Bioaerosol Spectrometer (fluorescence + morphology)	Norway (Svalbard, Zeppelin Observatory)
Kang et al.	Moisture-coupled aerosol filtration in moss-based porous media: enhanced particle capture and aerodynamic resistance trade-off	<i>Build Environ</i>	Metadata only (no abstract)	Metadata only - filtration efficiency vs pressure drop in moss media under humidity (by title)	South Korea (by author affiliation; not asserted)
Mathissen et al.	Development of the Worldwide Harmonized Heavy Duty Brake Cycle (WHBC) for Brake Wear Particle Measurement	<i>Atmosphere</i>	Test-cycle development + dynamometer testing	Brake-wear particles (PM ₁₀ /PM _{2.5} , PN), heavy-duty vehicles	Global / multi-region (dataset)
Morel et al.	Evaluation of Two Cyclonic Bioaerosol Samplers for Hospital Air Monitoring in a Calibrated Test Chamber Integrating Aerodynamics and Biological Endpoints	<i>Indoor Air</i>	Laboratory sampler validation	Bioaerosols (SARS-CoV-2, <i>S. aureus</i> , <i>A. fumigatus</i>); cyclonic samplers	France (test chamber)
Rawat et al.	Integrated Assessment of PM _{2.5} in the Central Himalaya using Satellite, Reanalysis, and a new In-situ CUPI Sensor network	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - satellite + reanalysis + a new in-situ low-cost PM sensor network (by title)	India (Central Himalaya)
Rosca & Stancu	Indirect Machine Learning Prediction of a Composite Indoor Air Quality Index in an Airport Passenger Transit Area	<i>Indoor Air</i>	Machine-learning model	Indoor PM _{2.5} , PM ₁₀ and CO ₂ sensor data; composite IAQ index	Italy (Brindisi Airport)

First author	Focus	Journal	Design	Metric	Region
Sahin et al.	Validation Design Governs the Apparent Performance of Machine Learning Calibration for Low-Cost Optical Particle Counters: A PM1 Study	<i>Atmosphere</i>	Multi-instrument field evaluation	Indoor PM1 from three low-cost OPCs vs reference nephelometers; six ML calibrators	Not stated (indoor co-location)
Tu et al.	Assessing the Need for Site-Specific Calibration of ATP Bioluminescence Sensors for Airborne Bacteria Monitoring in Indoor Environments	<i>Indoor Environ</i>	Metadata only (no abstract)	Metadata only - ATP bioluminescence sensors for airborne bacteria (by title)	Not recoverable from metadata
Wang et al.	Artificial intelligence for atmospheric chemistry: from observation to process understanding	<i>npj Clim Atmos Sci</i>	Metadata only (no abstract)	Metadata only - AI in atmospheric chemistry observation and process studies (by title)	Not recoverable from metadata
Xu et al.	Integrating Fragmented Mobile Monitoring Data into a High-Resolution Spatiotemporal Exposure Model for Black Carbon	<i>Environ Sci Technol</i>	Mobile monitoring campaign	Equivalent black carbon (eBC), vehicle-platform aethalometry	China (Beijing)
Yan et al.	Baseline Determination for Threshold-Based Interpretation and Groups of Single-Particle Aerosol Fluorescence Measurements	<i>Atmosphere</i>	Laboratory sampler validation	Single-particle fluorescence (SwisensPoleno Jupiter), bioaerosol 10-30 um	Ireland (laboratory + ambient case studies)
Yang et al.	An NB-IoT-Based Architecture with Spatial-Statistical Analytics for Cross-Domain Air and Water Quality Monitoring in Aquaculture	<i>Sensors (Basel)</i>	Real-time sensor campaign	Low-cost PM1/PM _{2.5} /PM ₁₀ channel among multi-pollutant air nodes	Taiwan (Tung-hai University facility)
Yang et al.	Development and Diagnosis of a Temporal-Context U-Net Deep Learning Surrogate Model for CMAQ Particulate Nitrate over China	<i>Environ Sci Technol Lett</i>	Machine-learning model	Hourly surface particulate nitrate, CMAQ emulation (July/December 2011-2020)	China (national)
Zhou et al.	Effects of geometrical parameters on eccentric cylindrical differential mobility analyzer	<i>Aerosol Sci Technol</i>	Metadata only (no abstract)	Metadata only - eccentric cylindrical DMA geometry (by title)	Not recoverable from metadata
Exposure assessment & modelling					
Akhtar et al.	Spatiotemporal characteristics of ozone and PM _{2.5} pollution in Jilin City: insights from observations in 2020 and 2023	<i>Environ Monit Assess</i>	Chemical transport model	Warm-season PM _{2.5} and O ₃ , 3 km WRF-CMAQ + surface monitors, meteorological	China (Jilin City)
Barthaux & Zuend	Estimating Speciated Input Concentrations for Biomass Burning Organic Aerosol Modeling	<i>Environ Sci: Atmos</i>	Metadata only (no abstract)	Metadata only - speciated BB VOC/POA model inputs (by deposit teaser)	Not recoverable from metadata
Bi et al.	Associations of street morphology and local wind environment with annual cumulative ambient PM _{2.5} exceedance: Evidence from Wuhan	<i>J Environ Manage</i>	Machine-learning model	Annual cumulative excess PM _{2.5} exposure per street; CatBoost-SHAP + PLS-PM	China (Wuhan)
Blond et al.	Boundary-Layer Height Outweighs Wind Speed in the Meteorological Control of Urban Air Pollution: Unique-Variance Attribution and a City Typology for Ten French Cities	<i>Preprints.org (preprint)</i>	Statistical model	Station PM _{2.5} , PM ₁₀ , NO ₂ , O ₃ vs boundary-layer height and wind	France (10 cities)
Bloomberg et al.	Associations of individual exposure to particulate matter with next-day physical activity and sedentary behaviour in older adults: a micro-longitudinal study	<i>medRxiv (preprint)</i>	Panel study	Personal air-quality monitors (waking hours, 5 days) vs area-level PM _{2.5} /PM ₁₀	UK
Chen et al.	Solid Waste Burning: An Overlooked Atmospheric Source of p-Phenylenediamine Antioxidants and Their Quinones	<i>Environ Sci Technol Lett</i>	Field measurement	PPDs and PPD-quinones in PM _{2.5} from open burning and incineration	China
Guo et al.	Atmospheric Environment of the Northern Tianshan Urban Agglomeration, Xinjiang: A Comprehensive Review of Pollutant Characteristics, Photochemical Processes, and Health Implications	<i>Atmosphere</i>	Narrative review	PM _{2.5} , coarse dust, BC/BrC, O ₃ , VOCs (review)	China (Xinjiang)

First author	Focus	Journal	Design	Metric	Region
He B. et al.	Contrasting seasonal behavior and inhalation risks of atmospheric PAHs, PAEs, and OPEs in a heating-influenced urban atmosphere	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - particle/gas PAHs, phthalates, organophosphate esters (by title)	Not recoverable from metadata
He et al.	Interpretable machine learning for spatially differentiated nonlinear relationships between PM _{2.5} and landscape patterns in an urban-rural integrated zone	<i>J Environ Manage</i>	Machine-learning model	Gridded PM _{2.5} vs landscape metrics (MSPA, FRAGSTATS, building height, NDVI)	China (Xuchang, Henan)
Kuntz et al.	Dynamics of tire-related chemicals in roadside aerosol over one year in Leipzig, Germany-influencing factors, potential source materials and comparison with rural aerosol	<i>J Hazard Mater</i>	Measurement campaign	Roadside PM ₁₀ filters; 40 tire-related chemicals by LC-MS/MS	Germany (Leipzig)
Liu et al.	Persistent cold air pools and associated sustained haze pollution in Taiyuan Basin	<i>J Environ Sci (China)</i>	Monitoring-record analysis	Surface PM _{2.5} vs basin meteorology (soundings, valley heat deficit, WRF)	China (Taiyuan Basin)
Liu et al.	Metropolitan atmospheric microplastics: Characteristics and its inputs to fluvial system	<i>J Environ Sci (China)</i>	Field measurement	Atmospheric microplastic deposition (n/m ² /day), fibres and fragments	China (Chengdu)
Luo et al.	Primary nitrated phenol emissions from heavy-duty diesel vehicles based on real-world activity data in Hunan, China	<i>Environ Pollut</i>	Modelling / inventory	Nitrated phenols (light-absorbing BrC), filter + PUF exhaust sampling	China (Hunan)
Ma M. et al.	Atmospheric Redistribution of Microplastics Reshapes Environmental Risk Across China	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - airborne microplastics (by title)	China (national, by title)
Mao et al.	Seasonal and Pollution-Level Variations in PM _{2.5} -Bound PFASs and Associated Health Risks in Jinan, a Typical Northern Chinese Industrial City	<i>Atmosphere</i>	Field measurement	PM _{2.5} -bound PFASs (17 compounds, LC-MS/MS), 163 filters	China (Jinan)
Mielnik et al.	Comparing geospatial interpolation techniques for estimating ambient PM _{2.5} mass concentrations from sugarcane burning in South Florida	<i>Research Square (preprint)</i>	Statistical model	Daily PM _{2.5} from a dense low-cost sensor network (46 sites), IDW vs kriging	USA (Florida, Everglades Agricultural Area)
Momen et al.	Explainable Machine Learning for Predicting Thermal Discomfort from Multipollutant Exposure on Dhaka Roadways	<i>Research Square (preprint)</i>	Field measurement	Roadside PM _{2.5} with noise and thermal sensation votes, monsoon	Bangladesh (Dhaka)
Munteanu et al.	Evaluating Graph-Centrality Context for Station-Level Correction of CAMS Air Quality Reanalysis in Lombardy from 2018 to 2021	<i>Sensors (Basel)</i>	Machine-learning model	CAMS reanalysis PM _{2.5} , PM ₁₀ , NO _x , NO ₂ corrected to ARPA stations	Italy (Lombardy)
Na et al.	Satellite-based estimation of PM _{2.5} chemical composition under sparse observations in Seoul metropolitan area	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - satellite-derived PM _{2.5} composition (by title)	South Korea (Seoul, by title)
Nassir et al.	A multi-sensor satellite assessment of air quality during the 2026 Middle East Conflict across nine regional countries	<i>Environ Pollut</i>	Satellite time-series analysis	AOD (MODIS MAIAC), TROPOMI/OMI NO ₂ , SO ₂ , CO, CH ₄ ; CAMS PM _{2.5} (indeterminate)	Middle East (nine countries)
Niamien et al.	Spatio-temporal analysis of the physical properties of aerosols in urban and rural environments in Cote d'Ivoire	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - aerosol physical properties (by title)	Cote d'Ivoire (by title)
Oliveira et al.	Nanominerological and geochemical signatures of coal-derived particles in playground dust	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - coal-derived nanoparticles in settled dust (by title)	Not recoverable from metadata
Rawat et al.	Size-resolved aerosols from residential fuel combustion in the northwestern Himalayas: potential effects on light absorption and human health	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - size-resolved residential-combustion aerosol (by title)	India (northwestern Himalayas, by title)

First author	Focus	Journal	Design	Metric	Region
Shahadat et al.	Spatiotemporal variability, trends, and health risk assessment of ambient PM _{2.5} and PM ₁₀ in Bangladesh during 2018-2025.	<i>Environ Geochem Health</i>	Monitoring-record analysis	16 DoE continuous monitoring stations, PM _{2.5} and PM ₁₀ , 2018-2025	Bangladesh (national, 16 stations)
Sheidaei et al.	A Multi-Stage Deep Learning Framework for Daily PM _{2.5} Estimation at 100 m Resolution Across the Contiguous United States, 2000-2024	<i>Environ Sci Technol</i>	Machine-learning model	Daily PM _{2.5} surface, 100 m, from gap-filled MAIAC-type AOD + meteorology + geospatial predictors	USA (contiguous)
Shoukat et al.	Suspended Atmospheric Microplastics in the Outskirts of Seoul City: Seasonal Variability, Correlation with Meteorology	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - suspended airborne microplastics (by title)	Not recoverable from metadata
Singh et al.	Aerosol Absorption Coefficients Across Global Sites: Impacts of Chemical Composition, Emissions, and Meteorology	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - aerosol absorption coefficients at global sites (by title)	Not recoverable from metadata
Surdu et al.	Summertime Aerosol Composition in Southern Greenlandic Fjords: Impacts of Sources and New Particle Formation Events	<i>Environ Sci Technol</i>	Field campaign / shipborne	NEMS-FTIR aerosol composition from two research vessels; PMF; number size distributions	Greenland (Narsaq fjords)
Vukic et al.	Integrated assessment of road and maritime traffic impacts on urban air quality and noise in a port-city environment	<i>Environ Monit Assess</i>	Field measurement	Short-duration low-cost PM _{2.5} /PM ₁₀ and CO at 10 sites; vessel and vehicle counts	Croatia (Split)
Wang et al.	Organosulfate-driven amplification of secondary organic aerosol formation from isoprene oxidation intermediates: Coupled effects of SO ₂ and relative humidity	<i>J Environ Sci (China)</i>	Chamber / laboratory	SOA from methacrolein/OH photooxidation; organosulfates (GC-MS, ATR-FTIR)	Not stated (laboratory)
Wen and Chang	Estimating marine PM _{2.5} concentrations and exploring the drivers over the eastern China seas using multi-source data and machine learning	<i>Environ Pollut</i>	Machine-learning model	Marine PM _{2.5} from AOD + meteorology + NO ₂ /CO/O ₃ /SO ₂ , 2019-2024	China (eastern China seas)
Zheng et al.	Comparative analysis of multiple health risks from different air pollution events	<i>J Environ Sci (China)</i>	Measurement campaign	Size-resolved number 3 nm-10 µm, PM _{2.5} composition, OP; lung deposition model	China (Beijing)
Burden, policy & mitigation					
Bong et al.	Delayed Coal Plant Retirements in the Era of Data Centers Carry a Measurable Mortality Cost	<i>medRxiv (preprint)</i>	Exposure model (secondary data)	Coal-plant-attributable PM _{2.5} , plant-specific source-receptor modelling	USA (Medicare, 20 coal plants)
Busskamp et al.	A virtual limit value for ultrafine particles: Frequency of threshold exceedances across Germany based on data from the German Ultrafine Aerosol Network	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - UFP number-concentration threshold exceedances, GUAN (by title)	Germany
Cheong et al.	Community-level air quality and health implications of sustainable aviation fuels adoption: a scoping review	<i>J Expo Sci Environ Epidemiol</i>	Metadata only (no abstract)	Metadata only - aviation-fuel emissions incl. ultrafine particles (by title)	Not recoverable from metadata
Jin Z. et al.	Carbonaceous Components Underlying Wildfire Smoke PM _{2.5} -Associated Mortality in Older U.S. Adults	<i>Environ Sci Technol</i>	Cohort	Annual smoke vs non-smoke PM _{2.5} mass and carbonaceous fraction	USA (contiguous)
Lee et al.	Exhaust Emission Characteristics of Gasoline Engines with Secondary Air Systems and Electrically Heated Catalysts in Cold-Start WLTC Mode	<i>ACS Omega</i>	Engine dynamometer testing	Particle number (PN) with CO, THC, NO _x over the cold phase of the WLTC	South Korea (laboratory)

First author	Focus	Journal	Design	Metric	Region
Lorences Noguera & Cardenas-Duran	Disentangling Population Ageing from Epidemiological Change in the Rising Burden of PM _{2.5} -Attributable Non-Communicable Disease	<i>QJM</i>	Metadata only (no abstract)	Metadata only - PM _{2.5} -attributable NCD burden decomposition (by title)	Not recoverable from metadata
Ma et al.	Afforestation and Emission Cuts: Dual Drivers for China's Clean Air and Carbon Neutrality	<i>Environ Sci Technol</i>	Chemical transport model	Summer PM _{2.5} (SOA) and MDA8 O ₃ from BVOC and anthropogenic scenarios	China (national)
Manchanda et al.	Inverse Modeling Identifies Efficient Emission Control Strategy for Mitigating PM _{2.5} Pollution	<i>Environ Sci Technol</i>	Chemical transport model	Annual PM _{2.5} vs precursor emissions by sector and location	USA (contiguous)
Pannu et al.	Assessment of the significance of urban vegetation in air phytoremediation: Quantitative analysis of particulate matter removal	<i>Research Square (preprint)</i>	Field measurement	Leaf-deposited PM in 0.2-2.5, 2.5-11 and 11-100 um fractions; elemental composition	India (Delhi)
Quadros et al.	Global air quality and human health impacts of growing aircraft emissions	<i>Commun Earth Environ</i>	Health impact model	Aviation-attributable PM _{2.5} and O ₃ from a global CTM, 2019 and 2040 scenarios	Global
Wang et al.	What drives long-term variations in biomass burning emissions in China?	<i>J Environ Sci (China)</i>	Modelling / inventory	Biomass-burning PM _{2.5} , PM ₁₀ , EC, OC and gas emissions, 2008-2023	China (national, provincial)
Yalali et al.	Geospatial modeling of environmental and socioeconomic drivers of health outcomes.	<i>Front Public Health</i>	Spatial regression (GWR)	Census-tract PM _{2.5} among environmental and socioeconomic determinants	USA (Harris County, Texas)
Yang et al.	Process-resolved characterization and plant-level inventory of multi-pollutant emissions from China's coking industry	<i>Environ Res</i>	Modelling / inventory	Size-segregated stack PM, ions, trace elements; plant-level inventory 2012-2024, projections to 2050	China (national, 307 plants)
Zhang et al.	Impact of strong biomass burning in Northeast China: air pollution, enhancement ratio among pollutants, and health risk	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - biomass-burning episode, pollutant enhancement ratios, health risk (by title)	China (Northeast)
Other clinical endpoints					
Gyeltshen et al.	Short-Term PM _{2.5} Exposure and Daily Cancer Mortality Across 47 Japanese Prefectures: A Nationwide Time-Stratified Case-Crossover Study, 2013-2022	<i>Int J Cancer</i>	Time-stratified case-crossover	Daily prefecture PM _{2.5} (lag 0-2)	Japan (47 prefectures)
Lima et al.	Epigenetic Exposure Signatures Distinguish Early-Onset from Later-Onset Colorectal Cancer in a Racially Diverse US Cohort	<i>medRxiv (preprint)</i>	Observational - cross-sectional	Methylation risk score for coarse PM (tumour tissue), not measured exposure	USA (DANCE cohort)
Sandal et al.	Disasters and hazard exposure in end-stage kidney disease: a scoping review of patient outcomes	<i>Clin Kidney J</i>	Scoping review	Poor air quality among heat, disaster and conflict hazards (metric varies by study)	Global (30 studies, 10 countries/regions)
Xia et al.	Long-term exposure to fine particulate matter constituents and lung cancer risk in Chinese never-smokers	<i>J Hazard Mater</i>	Prospective cohort	Annual PM _{2.5} and constituents (NO ₃ -, NH ₄ +, Cl-, SO ₄ 2-, BC, OM), ChinaHighAirPollutants 2004-2018	China (China Kadoorie Biobank, 10 regions)

First author	Focus	Journal	Design	Metric	Region
Yang and Liu	No evidence for large causal effects of six air pollutants on non-suppurative otitis media: a two-sample Mendelian randomization study	<i>medRxiv (preprint)</i>	Two-sample Mendelian randomisation	Genetic instruments for $PM_{2.5}$, $PM_{2.5}$ absorbance, $PM_{2.5-10}$, PM_{10} , NO_2 , NO_x (UK Biobank GWAS)	Europe (UK Biobank exposure; FinnGen R13 outcome)

Provenance and method

Period **27 September – 3 October 2026**, Sunday to Saturday, **seven daily issues, 125 records, 125 unique DOIs**, no record without a DOI. This rollup is **aggregation only**: it re-reads `state/corpus/*.json`, `state/metrics.csv` and `state/trials.json` over the period and **queries no literature API**; every record was gated by `check_dois.py` at daily-issue time. Tier mix: **16 A, 55 B, 54 C**. The W40 weekly was due on the 3 October run. That run built the 2 October daily and deferred both; the runs of 6, 7, 8 and 9 October then harvested without authoring, three of them halting mid-write. The 3 October daily and this rollup were built on the **9 October recovery run**, which also authored the 4–8 October dailies. The rollup includes the 3 October issue. One record admitted to the 3 October issue (10.1021/acs.est.6c07515) could not be authored on that run and is **held, not rejected**; it is absent from this pooled corpus and will enter a later issue. Effect estimates are transcribed verbatim from the daily issues on heterogeneous contrasts and are **not** meta-analysed. Abstracts remain the copyright of their respective publishers.